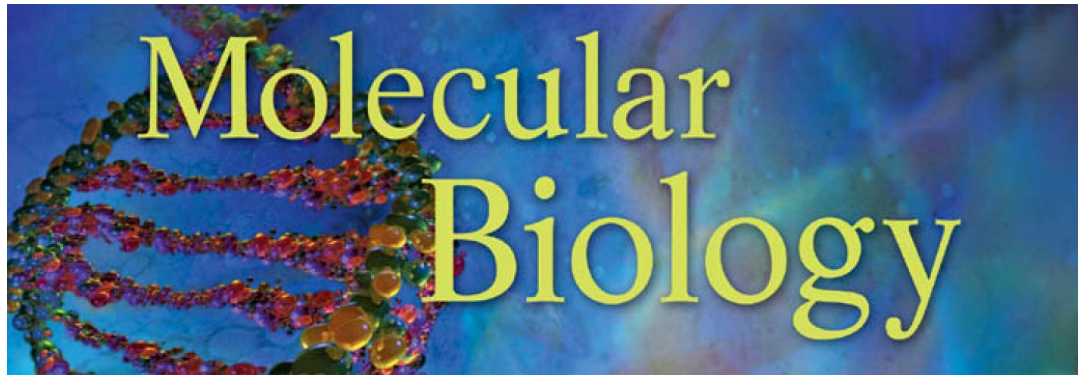




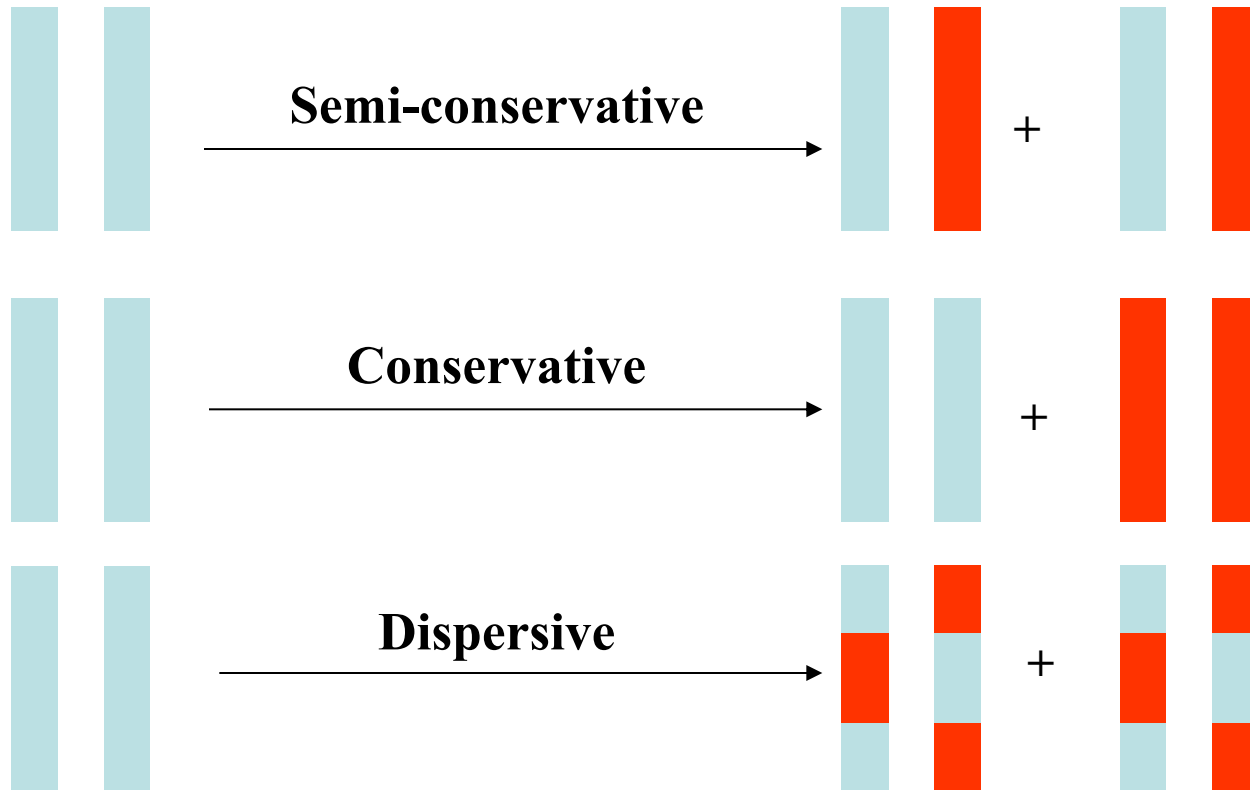
DNA replication

周明

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1. Semi-conservative



Three proposed ways for a double-stranded DNA molecule to be replicated. The red bar represents new synthesized DNA and the green bar represents parental DNA.

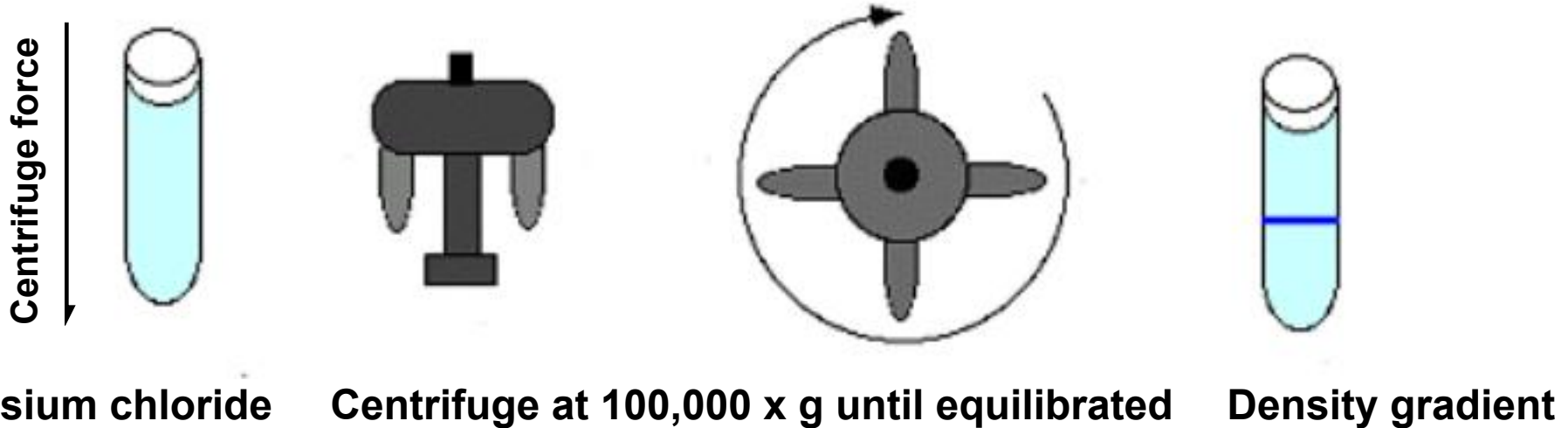
最初提出的三种可能的DNA复制模式。红色代表新合成DNA分子，绿色代表模板DNA分子。

The Meselson-Stahl Experiment

Different replication ways can produce different DNA progenies. Therefore, if the progeny DNA could be separated from parental DNA, the replication model can be determined. Based on this idea, Matthew Meselson and Franklin Stahl used ^{15}N isotope to create replicated DNA molecules with different weights, then ultracentrifuged to separate the DNA progenies

不同的复制方式产生不同的子代DNA分子。因此，如能把新合成的DNA分子从模板DNA分子分离开，则复制模式就能确定。根据这种思想，Matthew Meselson and Franklin Stahl设计用 ^{15}N 同位素产生不同重量的DNA分子及超速离心机分离不同重量DNA分子的实验。

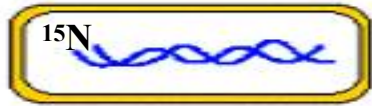
Density gradient centrifugation (密度梯度离心)



In equilibrium centrifugation, the density gradient extends to a density higher than that of one or more components of a mixture so that these components come to the equilibrium at a point equal to their own density and stop moving

用一定的介质在离心管内形成一连续或不连续的密度梯度，将细胞混悬液或匀浆置于介质的顶部，通过重力或离心力场的作用使细胞分层、分离。

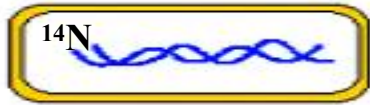
Mechanism of the Meselson-Stahl Experiment (1958)



E. coli grown many generations in ^{15}N medium (Heavy)



Then



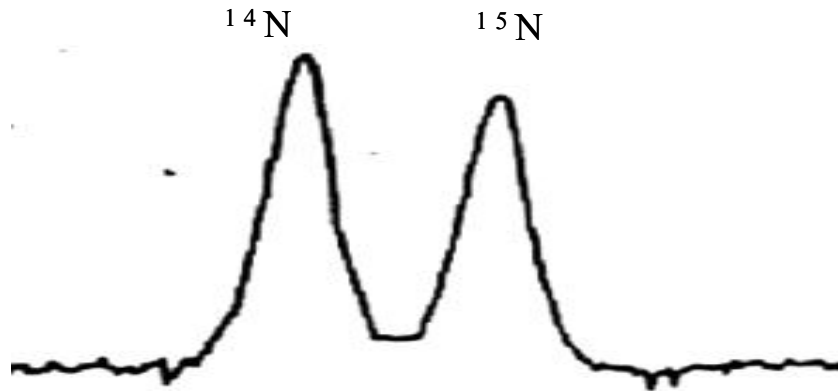
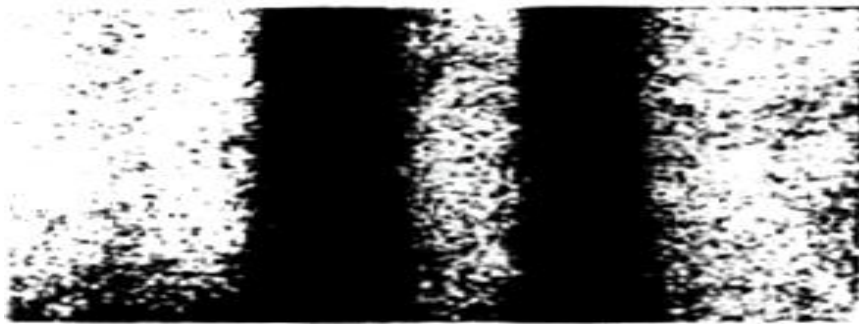
Grow in ^{14}N medium (light) and sampled after various generations



Extract DNA from cell and carry out “density gradient centrifugation”

大肠杆菌细胞在 ^{15}N 培养基（重培养基）中培养数代，产生含重氮（ ^{15}N ）的大分子（包括 ^{15}N -DNA分子），而后移到正常培养基（ ^{14}N ）上培养，在不同时间提取DNA样品并在CsCl溶液中超速离心。CsCl溶液在超速离心下形成密度梯度。DNA分子在该密度梯度中迁移，并在与其密度与溶液密度相同点达平衡。

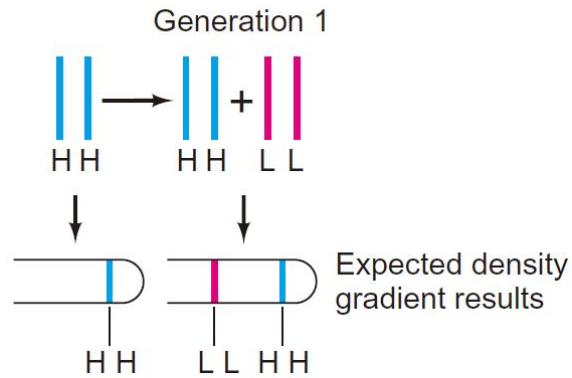
Methods



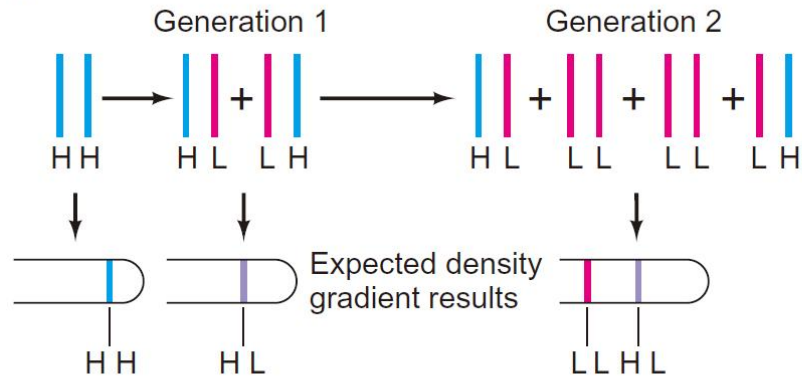
A mixture of ^{14}N and ^{15}N bacterial lysates, each containing about 10^4 lysed cells, was centrifuged in CsCl solution. The photograph was taken after 24 hours of centrifugation at 44,770 rpm. B: relative amounts of the two kinds of DNA by quantification of ultraviolet light absorption.

含 10^4 个 ^{14}N 与 ^{15}N 混合溶解细胞的菌液在CsCl溶液中经44, 770 rpm离心24小时。

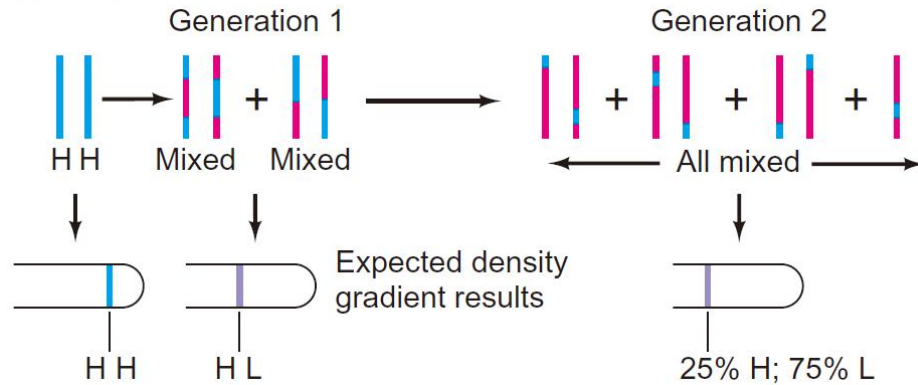
(a) Conservative

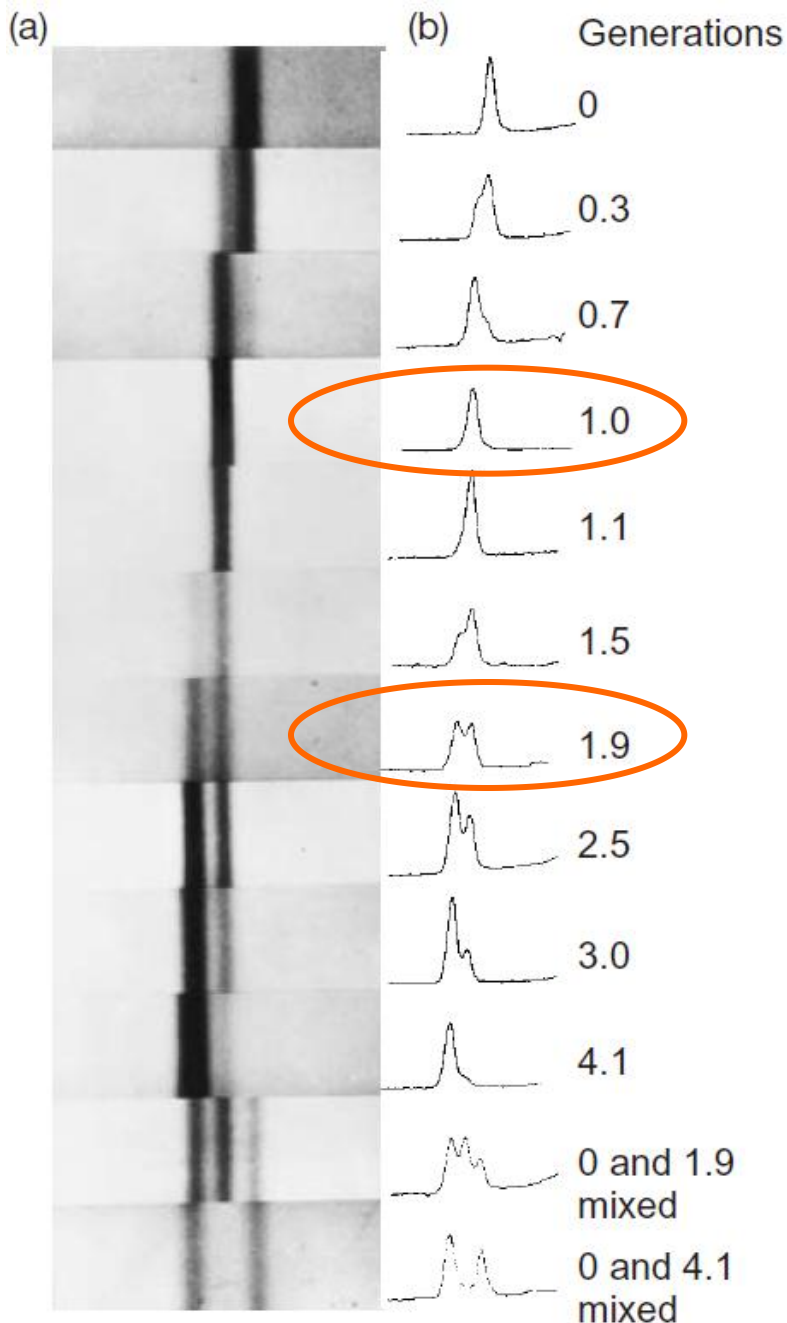


(b) Semiconservative



(c) Dispersive

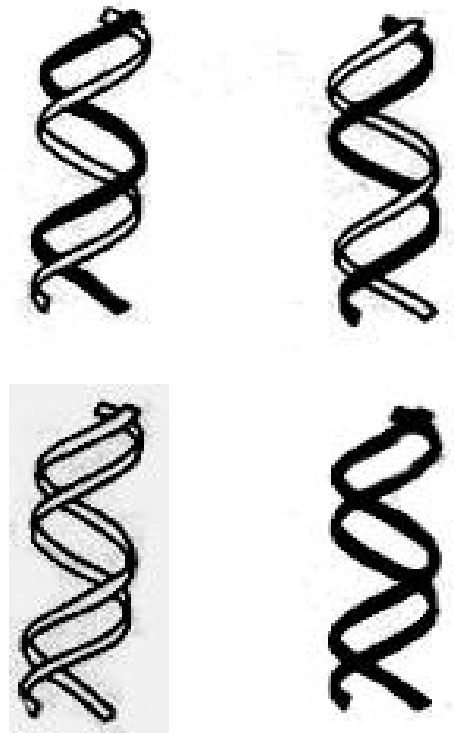
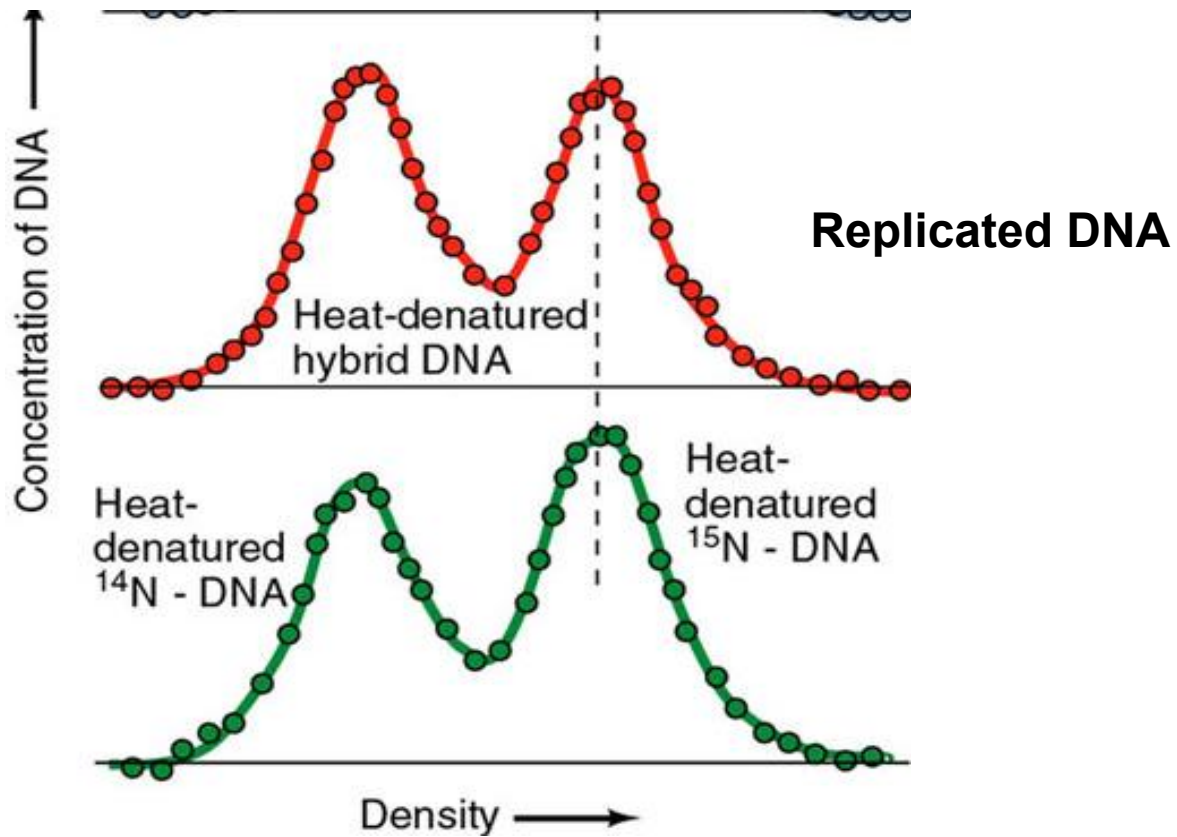




Results of the experiment

The results indicated that DNA replication is not conservative, because if the conservative model were correct, the first generation DNA would contain two bands, one light (^{14}N) and one heavy (^{15}N). Instead, it has only one band, of intermediate density.

结果表明，DNA复制不可能是保留复制。因为保留复制产生的第一代DNA离心后将会有两条带：一条“轻带”（ ^{14}N ），一条“重带”（ ^{15}N ），但实验结果表明第一代DNA只有一条具中等密度的带。



Double check

Summary

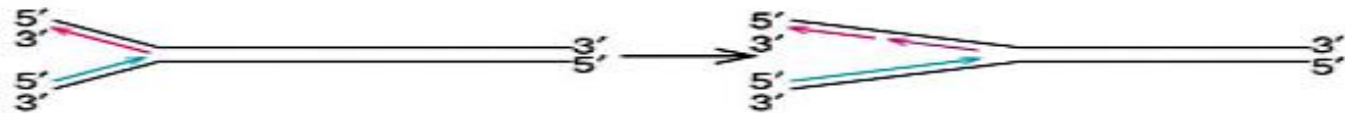
- Ultracentrifugation in CsCl solution of the individual DNA strands of the first generation samples and control samples. The results indicate that the hybrid DNA from the first generation after the shift has two bands that are of heavy and light density. Therefore, the dispersive model is ruled out and the conclusion is that DNA replicates by a semi-conservative mechanism.
- 复制第一代单链DNA及对照单链DNA在CsCl溶液中经超速离心密度梯度分离结果。结果表明，杂合DNA只有两条DNA带：一条轻带（ ^{14}N -DNA），一条重带（ ^{15}N -DNA）。因此，杂合DNA是嵌合型DNA的可能性被排除。DNA复制是半保留复制得到完全证明。

2. Semi-discontinuous replication

Continuous



Semi-discontinuous



Discontinuous



The 3 DNA synthesis models proposed by Okazaki.

Okazaki假设DNA合成有3种方式：连续、半不连续与不连续方式。

DNA polymerase & Okazaki's model

- The discovery of DNA polymerase by Arthur Kornberg has revealed that DNA polymerase synthesize DNA in one direction only: 5' to 3'.

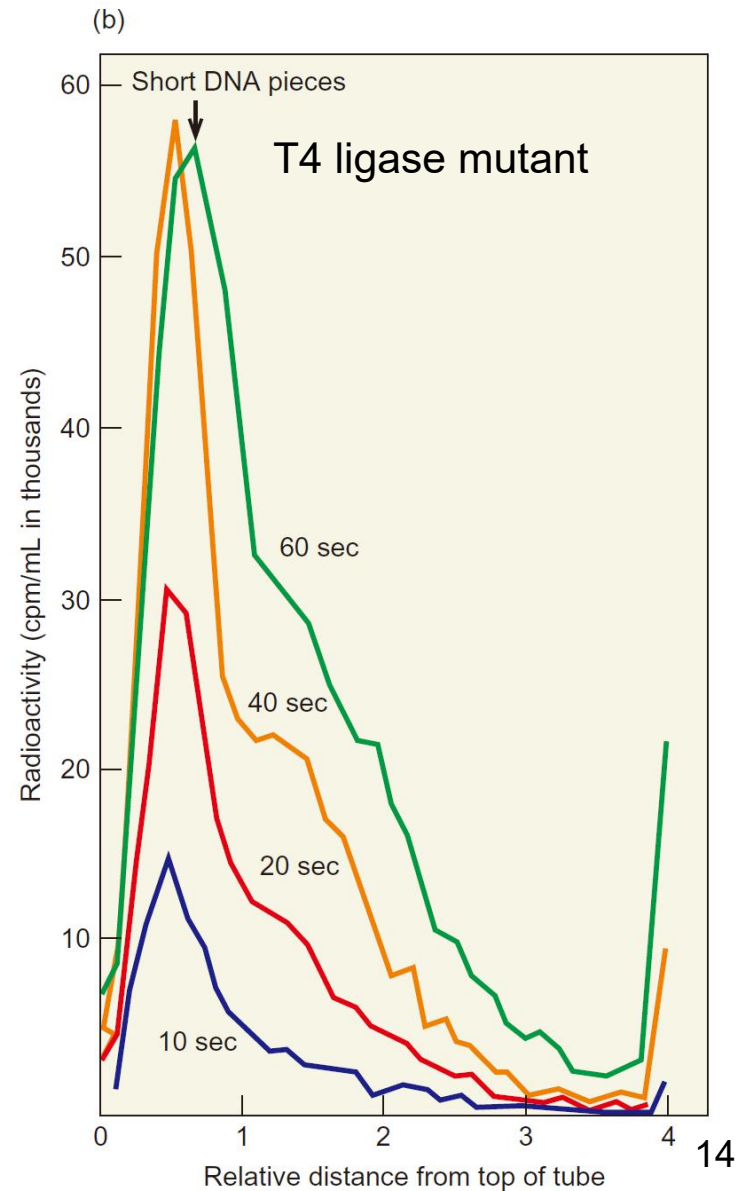
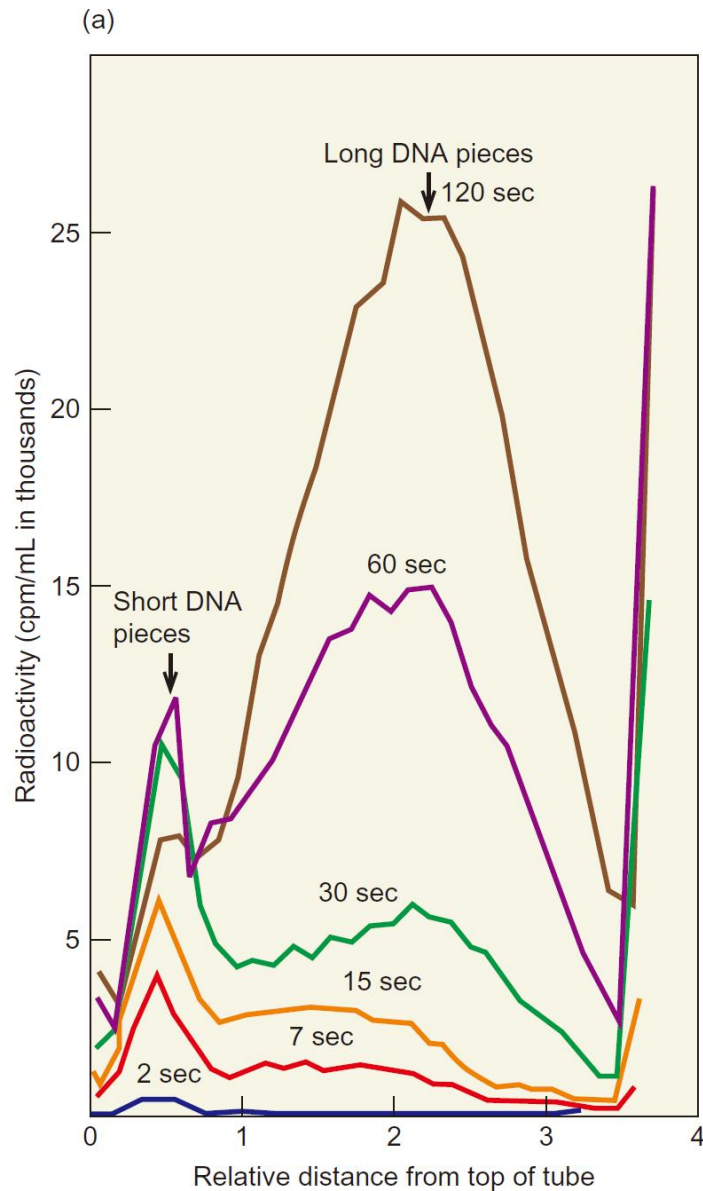
(Arthur Kornberg发现了DNA聚合酶并证明了该酶严格按模板3'-5'方向合成DNA, 1959 Nobel Prize).

- Okazaki proposed “Semi-discontinuous model”
- Okazaki fragment : DNA replication is Semi-discontinuous. One strand is replicated continuously in the direction of the movement of the replicating fork; the other is replicated discontinuously as 1-2 kb Okazaki fragments in the opposite direction. This allows both strands to be replicated in the 5' to 3'.

Discovery of Okazaki fragment

- Materials and Methods:
 1. Phage T4: simplicity; availability of T4 ligase mutant
 2. Shorter pulses of ^3H -labeled thymidine of *E.coli* cells that were replicating T4 DNA
 3. Using ultracentrifugation to measure the sizes of the newly synthesized DNAs

Discovery of Okazaki fragment



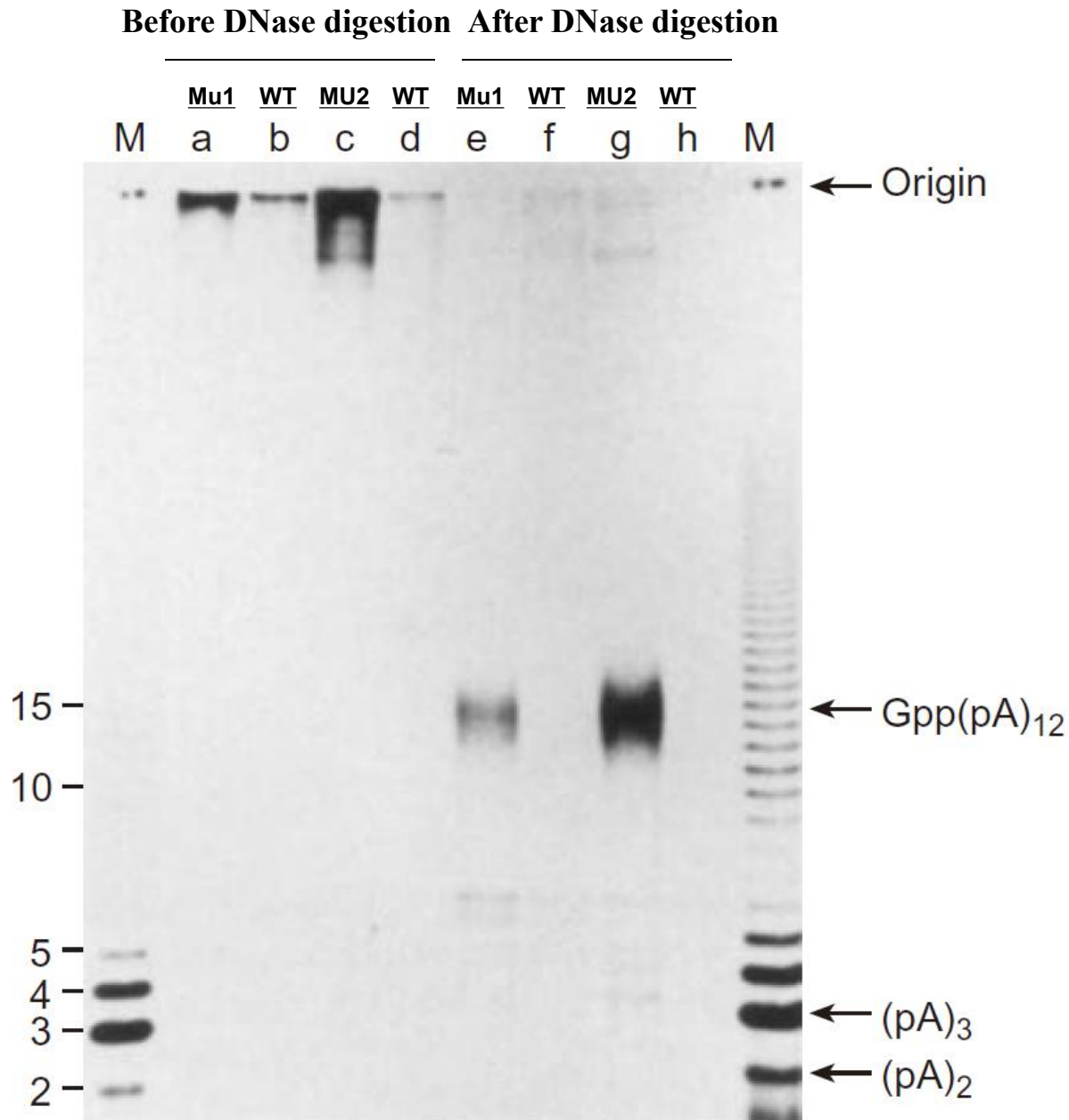
3. Primer dependence

Hypothesis:

DNA polymerase can not initiate DNA synthesis without a 10-12 nt long RNA primer

Evidence :

- Rifampicin inhibits M13 phage (*E. coil*) DNA replication (Rifampicin is known to inhibit RNA polymerase, not DNA polymerase)
- DNase can't completely destroy Okazaki fragment. It will left little piece of RNA 10-12 bases long



**Finding and measuring
RNA primers:**

**Mu1 and Mu2 are
mutant *E. Coli* cells
lacking one or both of
nucleases (RNase H or
exonuclease activity of
DNA polymerase) that
degrade RNA primers**

Features of DNA replication (DNA复制的特征)

1. Semi-conservative (半保留复制)
2. Semi-discontinuous (半不连续复制)
3. DNA polymerases are template- and primer-dependent
(DNA 聚合酶依赖于DNA模板与引物)
4. All DNA polymerases synthesize DNA in the 5' to 3' direction,
reading the template 3' to 5'
(所有的DNA聚合酶按5'-3'方向[模板DNA为3'-5'方向]合成DNA)
5. DNA polymerases require the four deoxynucleoside
triphosphates (dATP, dGTP, dCTP, and dTTP)
(DNA聚合酶合成DNA需要4种脱氧核糖核酸分子: dATP, dGTP, dCTP,
and dTTP)

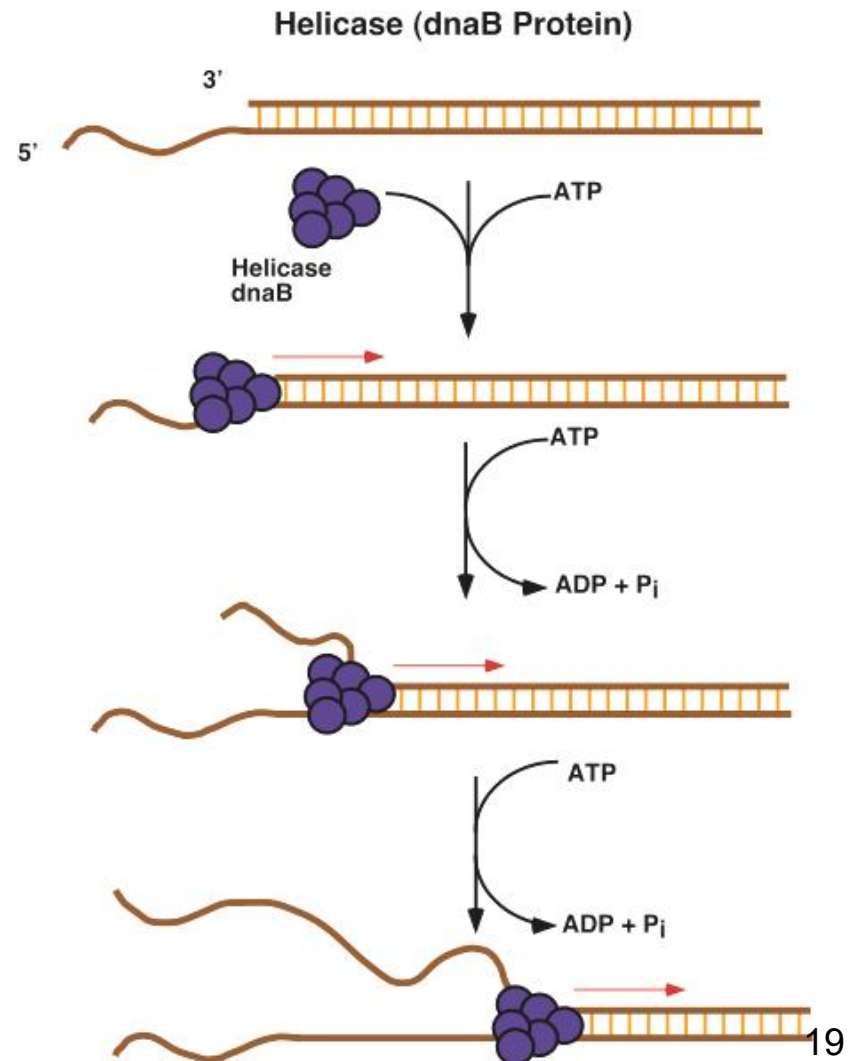
Enzymology of DNA replication

- Strand separation: Helicase
- Single-Strand DNA-Binding protein: SSBP
- Topoisomerases
- Primease
- DNA polymerases: three types
- DNA ligase
- Telomerase

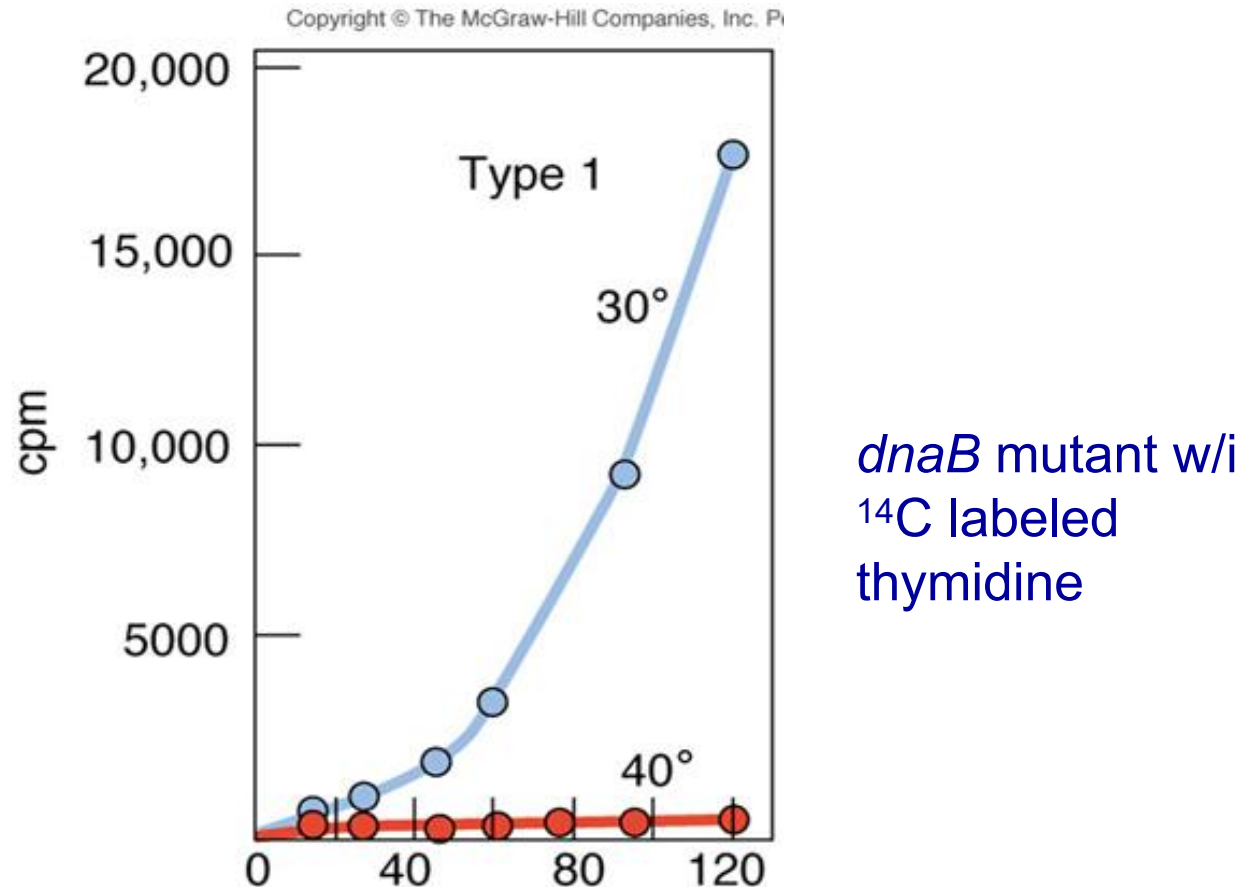
DNA Helicase (DNA解旋酶)

DNA helicase: the enzyme that harnesses the chemical energy of ATP to separate two parental DNA strands at the replicating fork.

There are about 10 proteins in *E.coli* that have helicase activity. The one involved in DNA replication is the product of the *dnaB* gene.

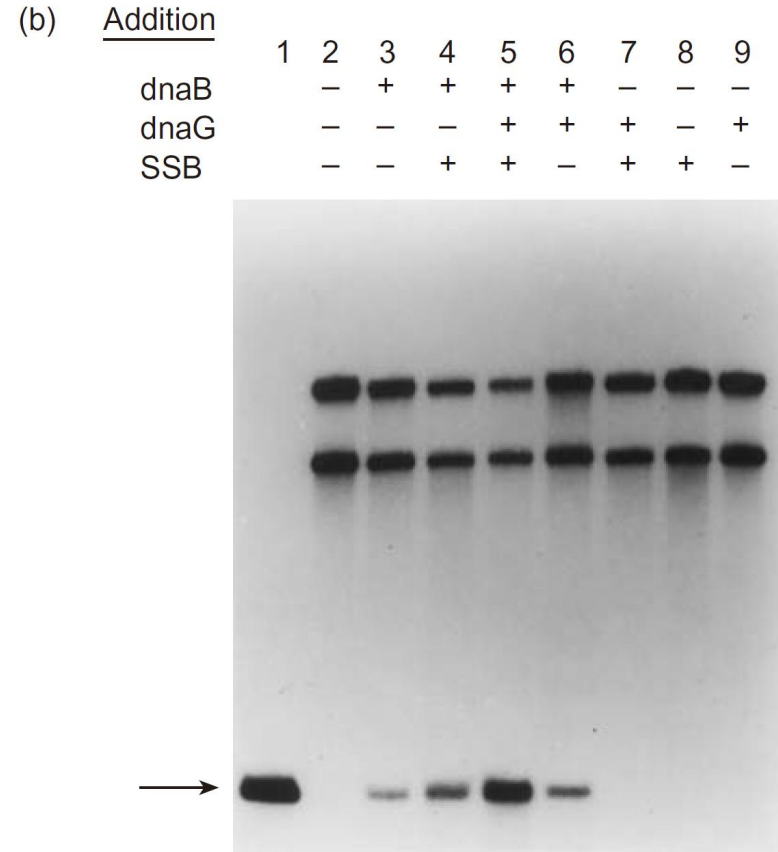
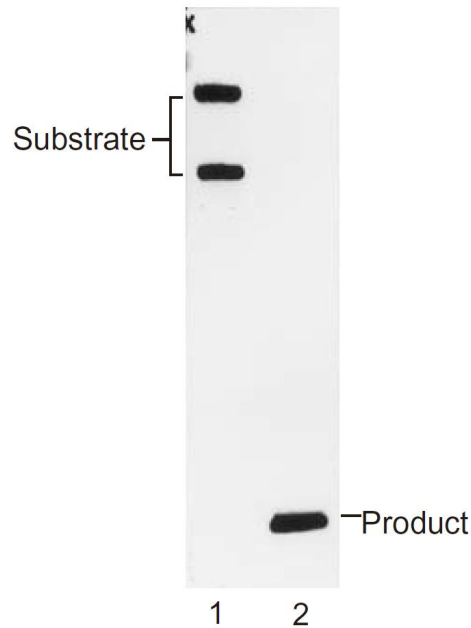
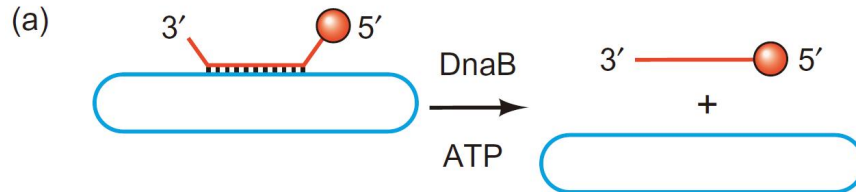


DnaB is essential for DNA replication



Temperature sensitive mutants

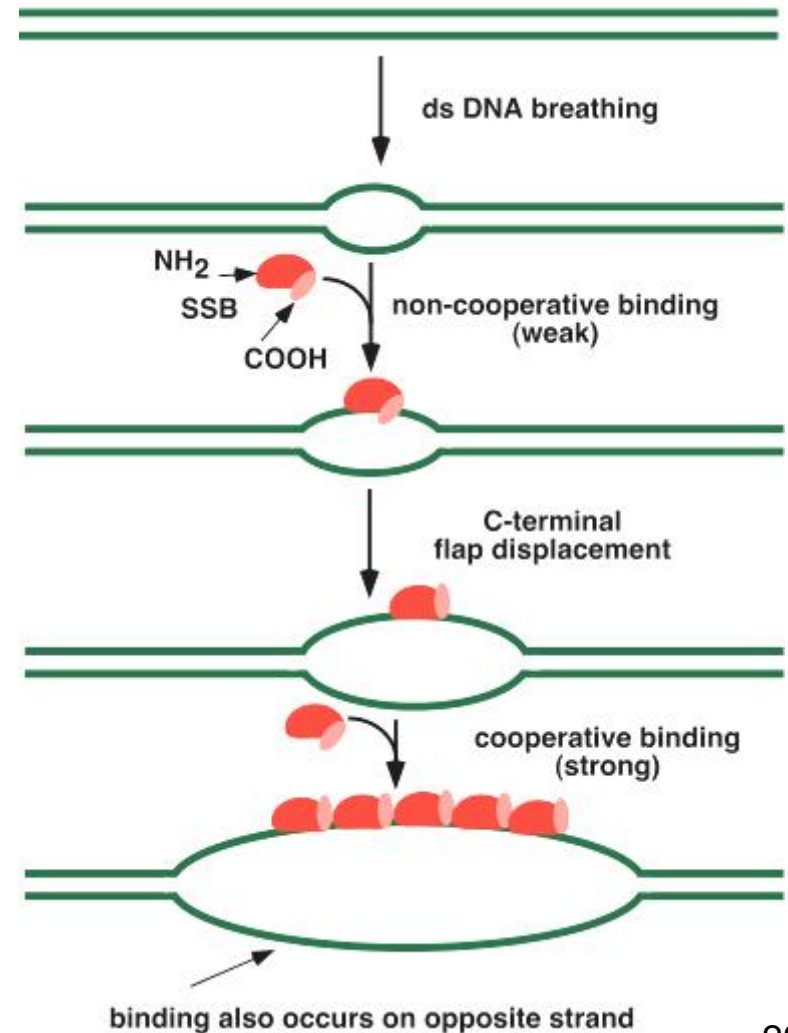
DnaB is a helicase



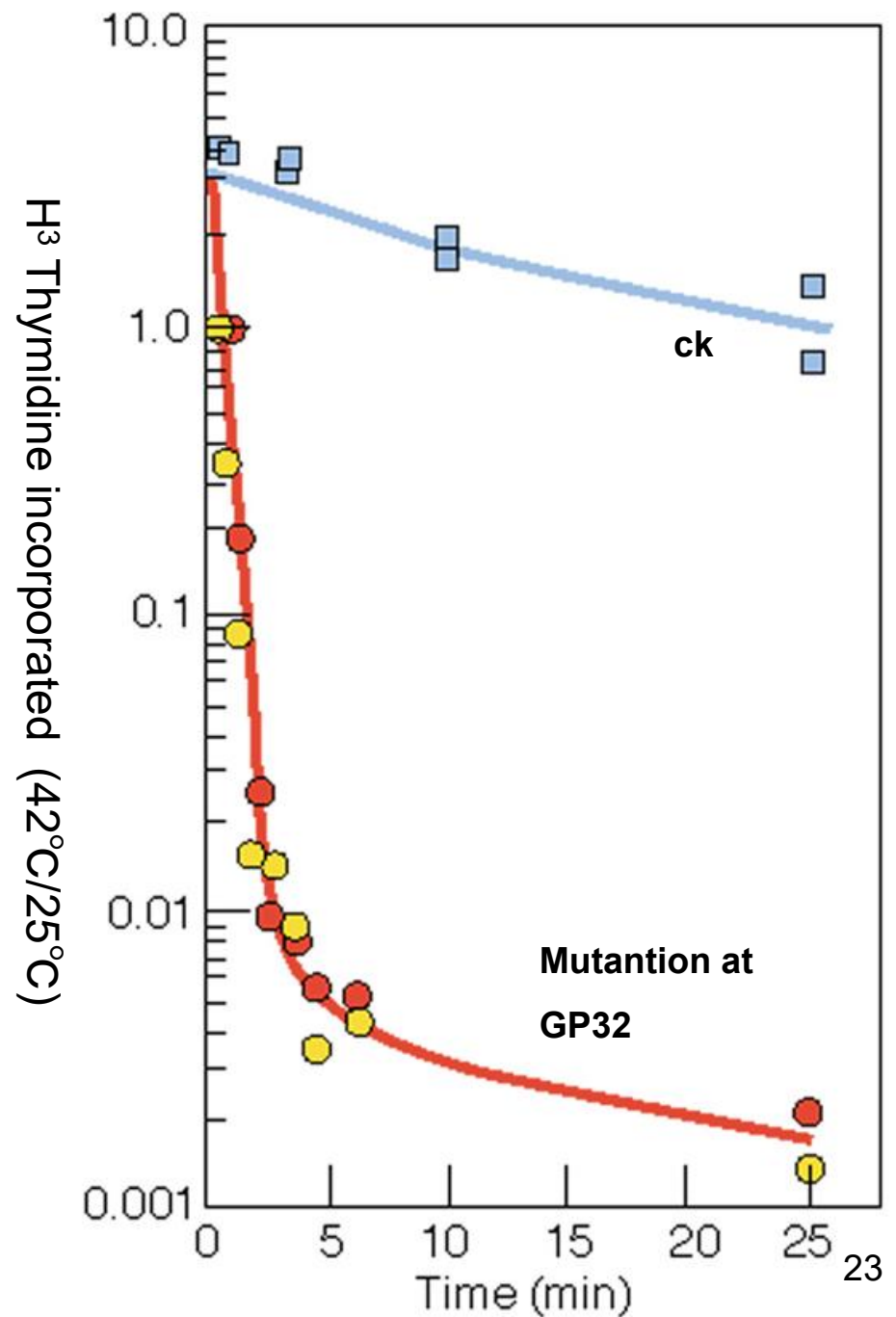
- DnaB has helicase activity on its own (lane3)
- The other proteins stimulate this activity (lane 4-6)
- w/o DnaB, no helicase activity (lane 7-9)

Single Strand DNA Binding Protein (SSBP, 单链结合蛋白)

SSBP: to keep the single stranded DNA separated and promoting the progress of the replication enzymes.

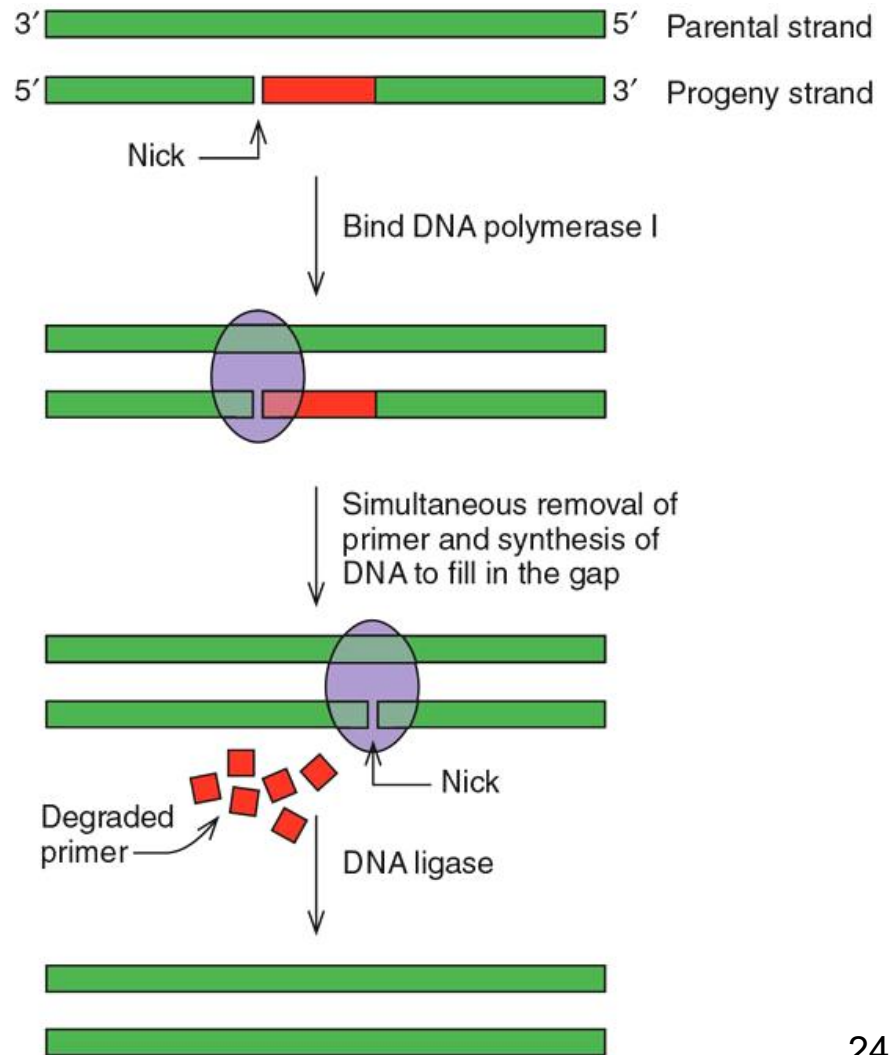


GP32 (SSB protein in T4 phage) is essential for DNA replication

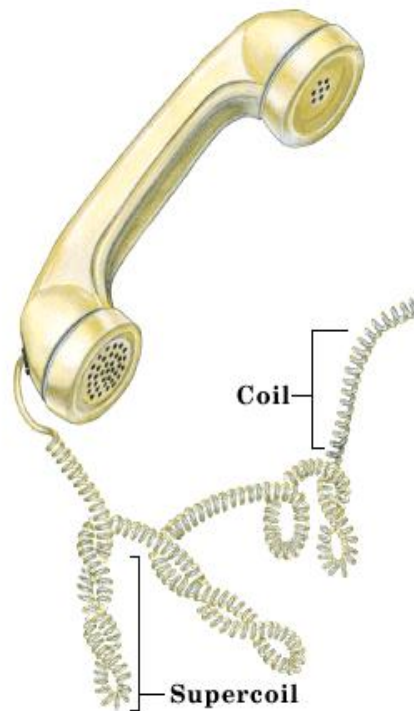


DNA Ligase (DNA连接酶)

- **DNA ligase:** Join fragments of DNA together.
- This takes place when the Okazaki fragments are ready to be linked together. First the RNA primer is removed and DNA synthesized in its place.



Topoisomerase (拓扑异构酶)



解链过程中正负超螺旋的形成

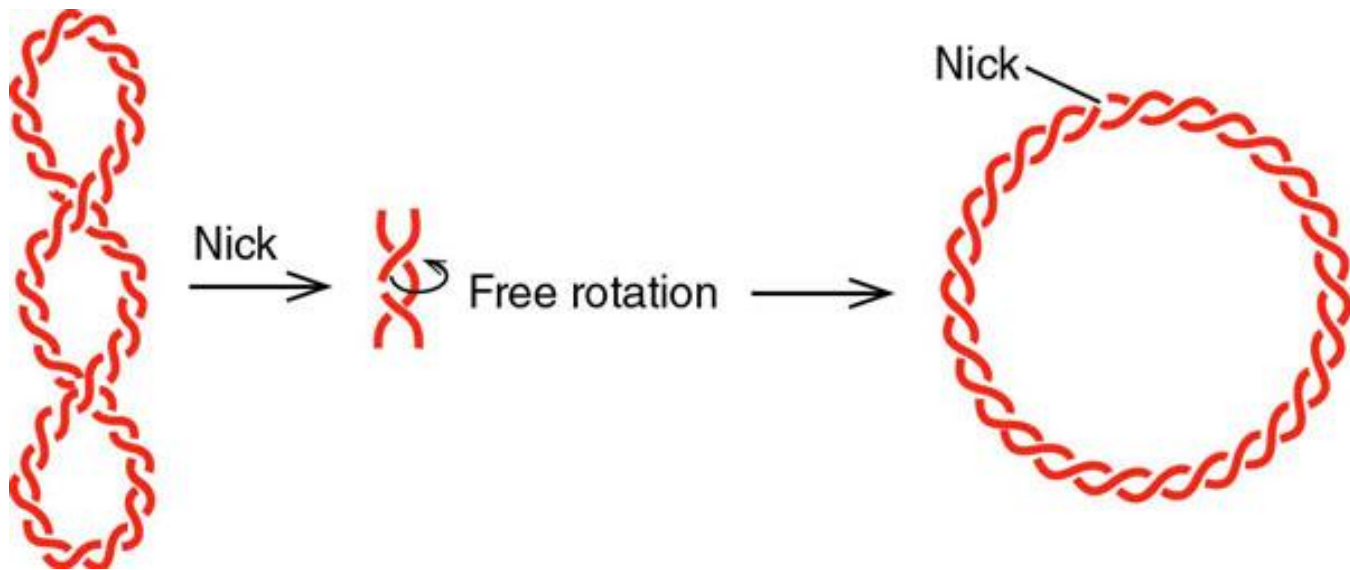


Topoisomerase (拓扑异构酶)

- Unwinding a helix for replication produces topological strain and supercoiling. This strain must be relieved or replication cannot continue. This is the function of enzymes called **topoisomerase**.
- Two types of topoisomerase:
 1. Introduce temporary single-strand breaks (nicks) called Type I topoisomerases
 2. Introduce temporary double-strand breaks, called Type II topoisomerases.

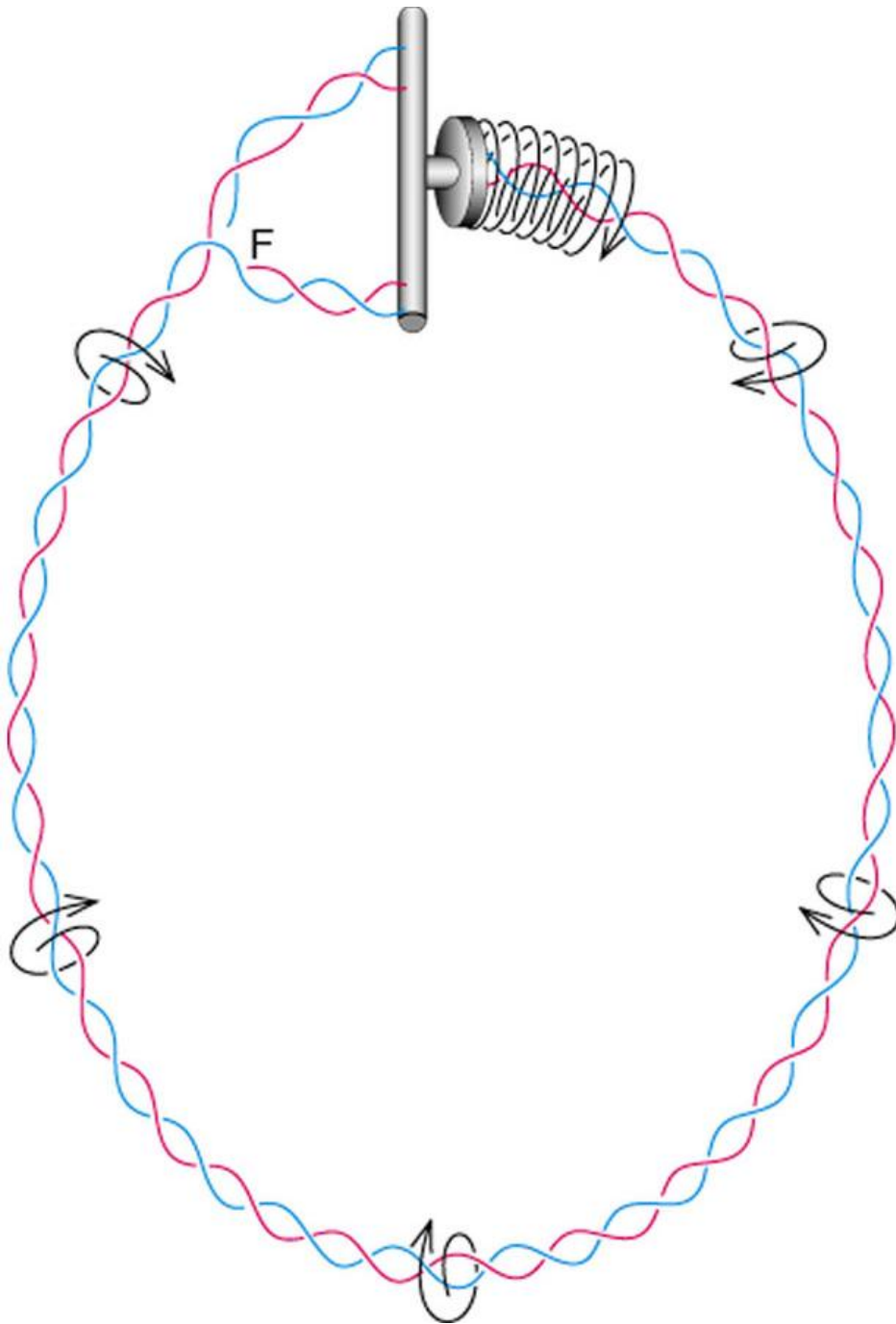
Type I topoisomerase

- These are also called nicking-closing enzymes and function to relax supercoiled circular molecules. These enzymes function without the input of energy in the form of ATP. The energy is provided by the release of strain in the molecule.
- 第一类拓扑异构酶也称为缺口闭合酶，参与DNA超螺旋结构的解除。该类酶促反应所需能量来自拓扑应力的释放。

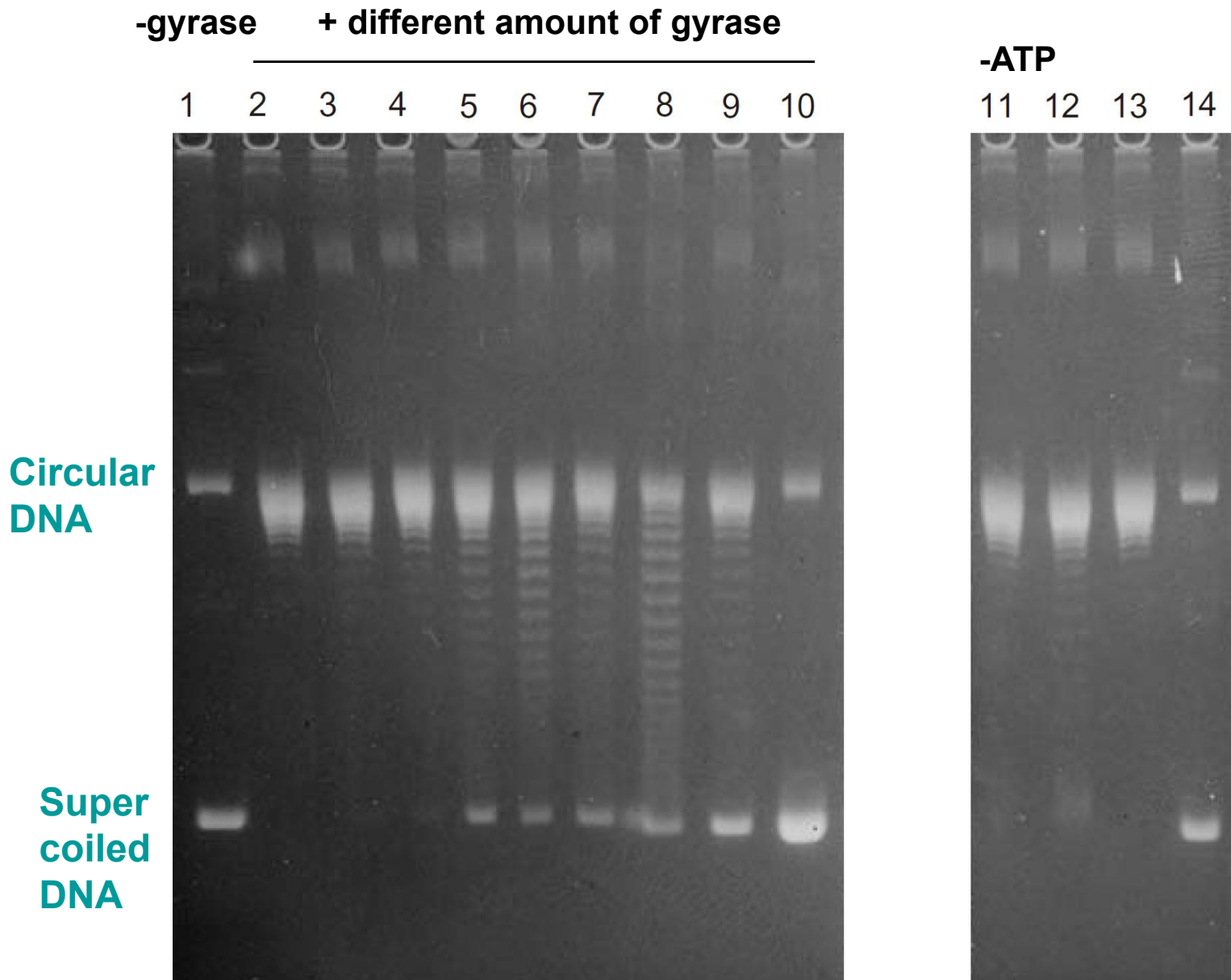


Type II topoisomerase

- Type II topoisomerases are also called DNA gyrases. These enzymes can either relieve supercoiling or introduce supercoiling into a DNA. They require the energy of ATP for their action
- 第二类拓扑异构酶也称为DNA旋转酶。这类酶即能解除超螺旋结构，也能产生超螺旋结构DNA。其酶促反应需要ATP提供能量。



DNA gyrase: a topoisomerase that pumps negative superhelical turns into DNA to relax the positive superhelical strain created by unwinding of DNA during replication



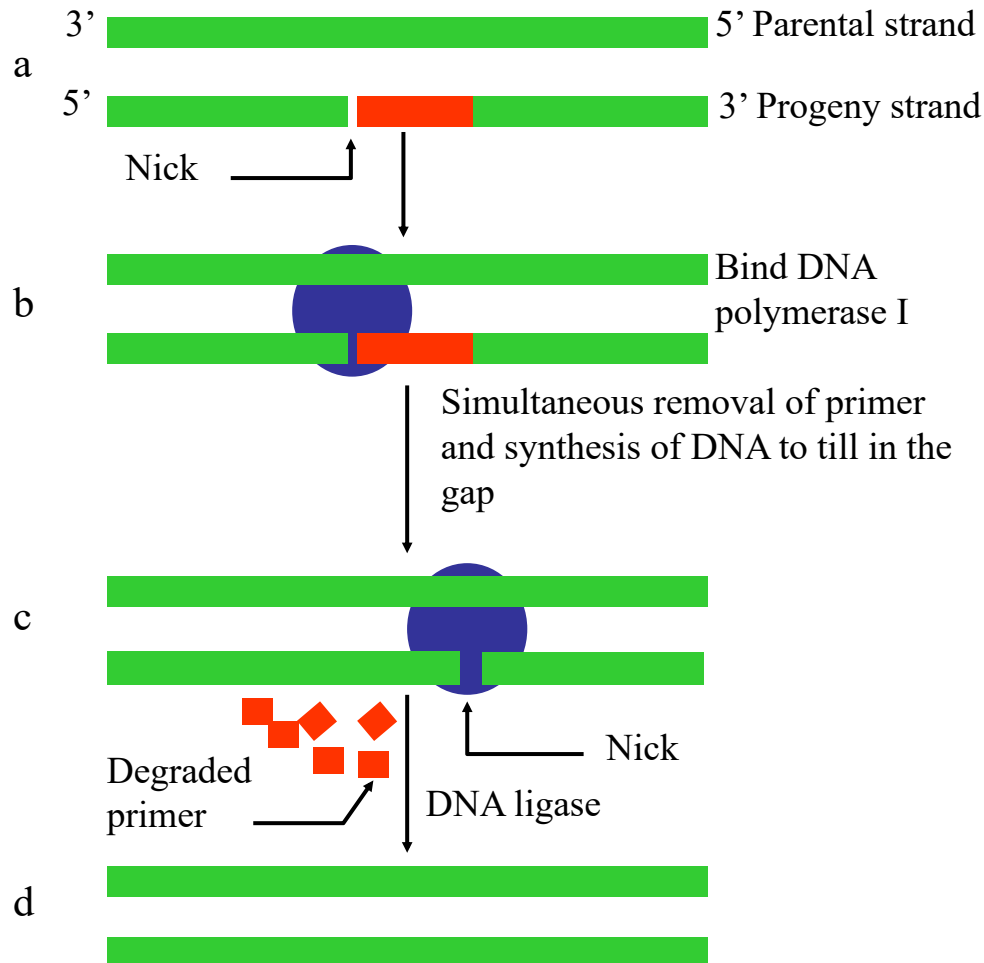
1: supercoiled ColE1 DNA; 2: - gyrase; 3-10: different amount of gyrase.

Prokaryotic DNA Polymerases

Characteristic	Polymerase I Kornberg polymerase	Polymerase II	Polymerase III
gene	<i>polA</i>	<i>polB</i>	<i>polC</i>
MW	103,000	90,000	130,000
# molecules/cell	400	100	10
3' exonuclease	yes (proofreading)	yes	yes
5' exonuclease	yes (remove RNA primer)	no	no
biological function	DNA repair, RNA primer excision	DNA repair	replicative chain growth

DNA polymerase I:

Removing of RNA primer, fill the gap that RNA primer leaves



DNA polymerase III holoenzyme

Table 20.1 Subunit Composition of *E. coli* DNA Polymerase III Holoenzyme

Subunit	Molecular mass (kD)	Function	Subassemblies				
α	129.9	DNA polymerase	Core	Pol III'	Pol III*	Pol III holoenzyme	
ϵ	27.5	3'→5' exonuclease					
θ	8.6	Stimulates ϵ exonuclease					
τ	71.1	Dimerizes core					
		Binds γ complex	γ complex (DNA-dependent ATPase)				
γ	47.5	Binds ATP					
δ	38.7	Binds to β					
δ'	36.9	Binds to γ and δ					
χ	16.6	Binds to SSB					
ψ	15.2	Binds to χ and γ					
β	40.6	Sliding clamp					

*Pol III holoenzyme minus the β -subunit.

Source: Reprinted from Herendee, D.R. and T.T. Kelly, DNA Polymerase III: Running rings around the fork *Cell* 84:6, 1996. Copyright © 1996, with permission from Elsevier.

The enzyme carries out the elongation of primers to make both the leading and lagging strands of DNA

Primase: Synthesis of the RNA primer

- RNA synthesis is not primer-dependent: the primer is synthesized by an enzyme called **RNA Primase or simply primase**.
- Primase is contained in a structure called the **primosome**. Primosome is composed of 6 proteins, α , α' , α'' , DnaB, C and I.
- The functional primosome is a complex of these 6 proteins and primase. Primosome primes the replication of lagging strand.

引物酶：RNA引物合成

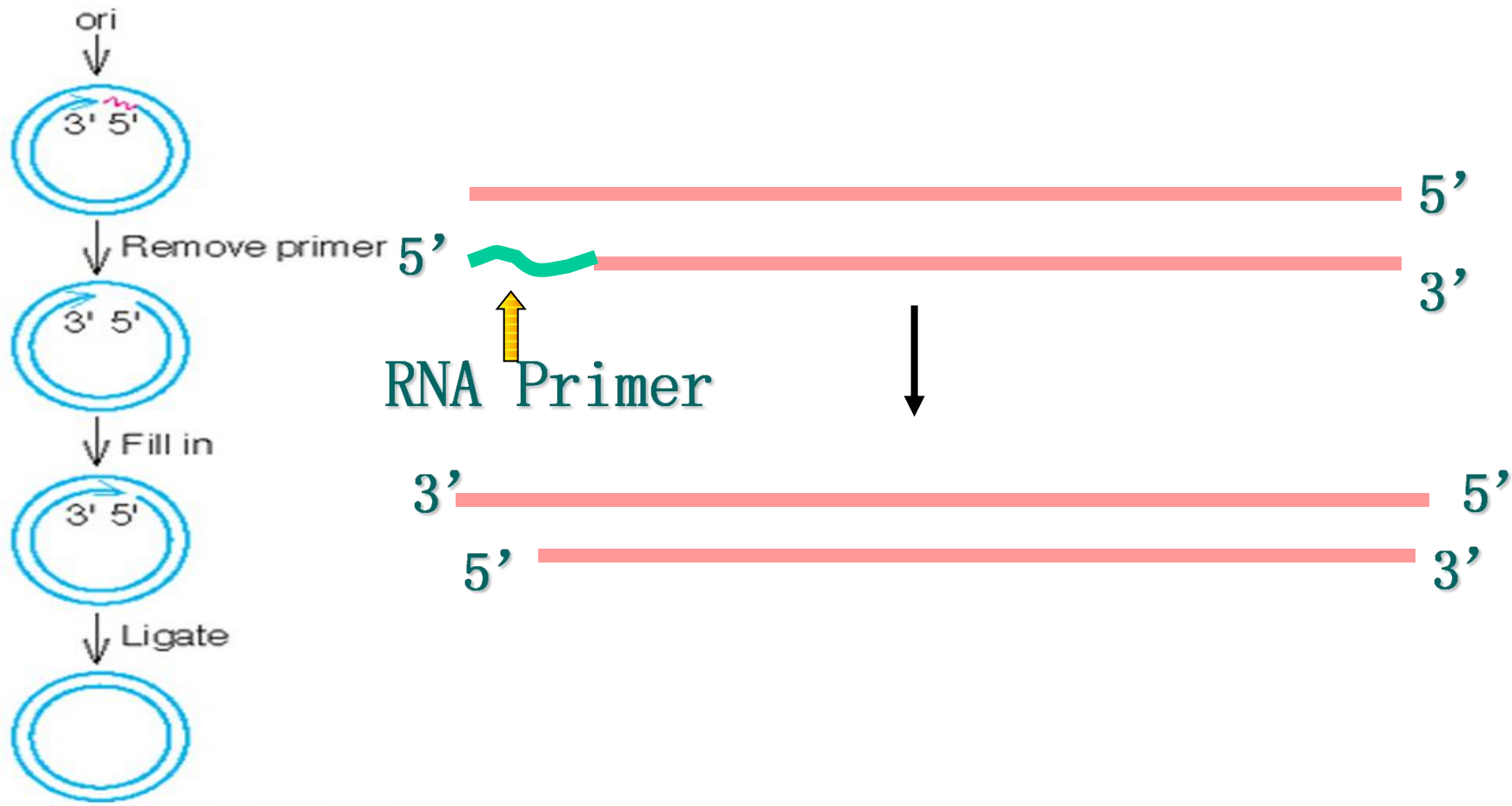
DNA复制的引物是RNA。引物RNA由RNA引物酶合成。因RNA合成无需引物，引物RNA的合成仅需以50个DNA碱基左右为模板。**引物酶**存在于**引物合成体（引发体）**。只有当引物合成体把6种蛋白, α , α' , α'' , DnaB, C与I及引物酶合在一起组装后，它才具功能。

Telomerases

- Problem: when a newly synthesized lagging strand has been copied near the 3' end of one strand. Now you need to have a primer to start another lagging strand. But how do you do this beyond the 3' end of the chromosome strand?
- **Telomerases:** The enzyme is related to reverse transcriptase, in that it has a small RNA template as a part of the enzyme structure.
- Telomeres have repeated sequences called telomere repeats. The RNA template of telomerase is used to add a telomere repeat to the 3' end. An RNA primer is then synthesized as a copy of this extension to the DNA and the last lagging strand is made. Later the extension is taken off along with the RNA primer.

Coping with the gaps left by primer removal

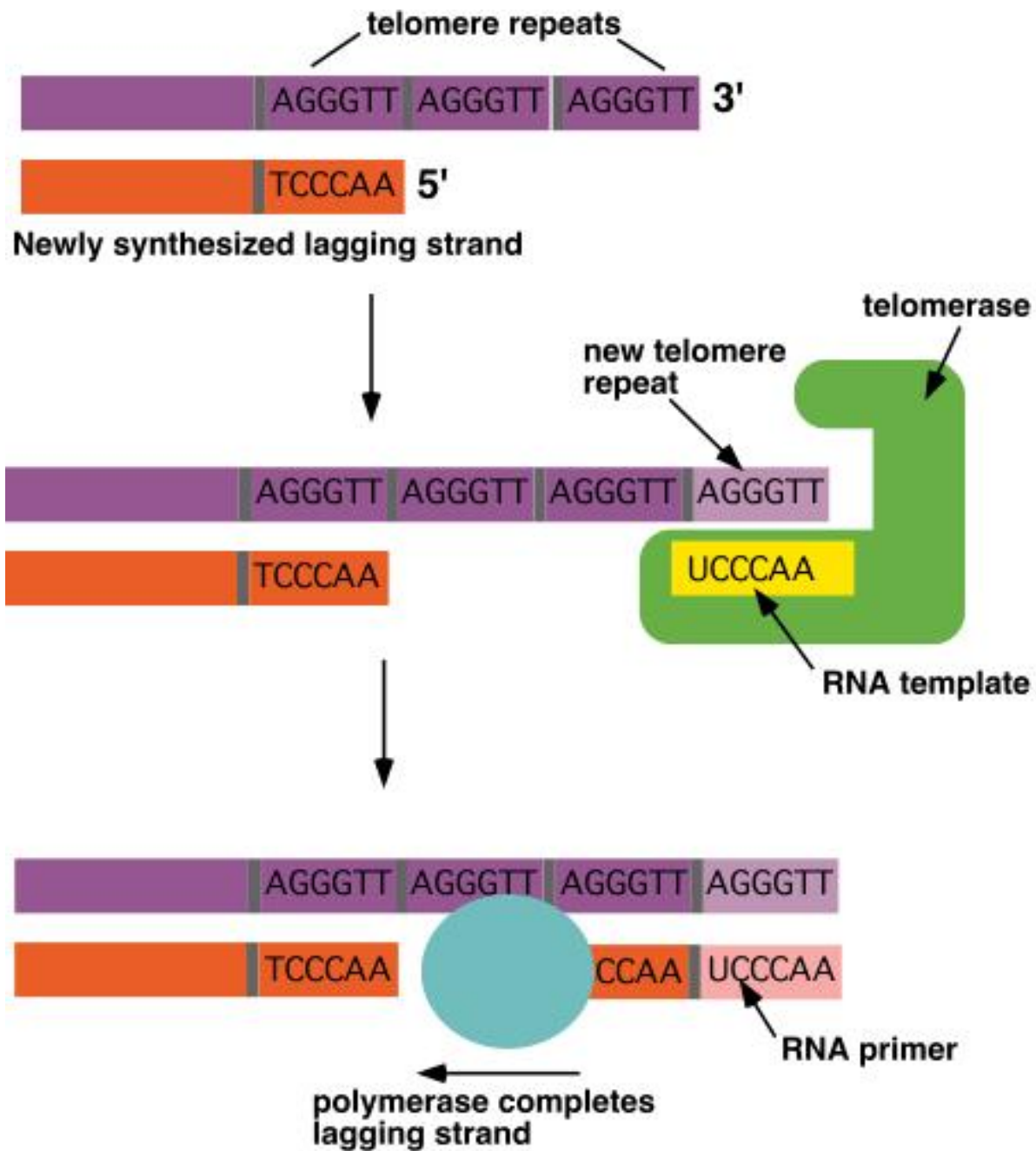
(a)

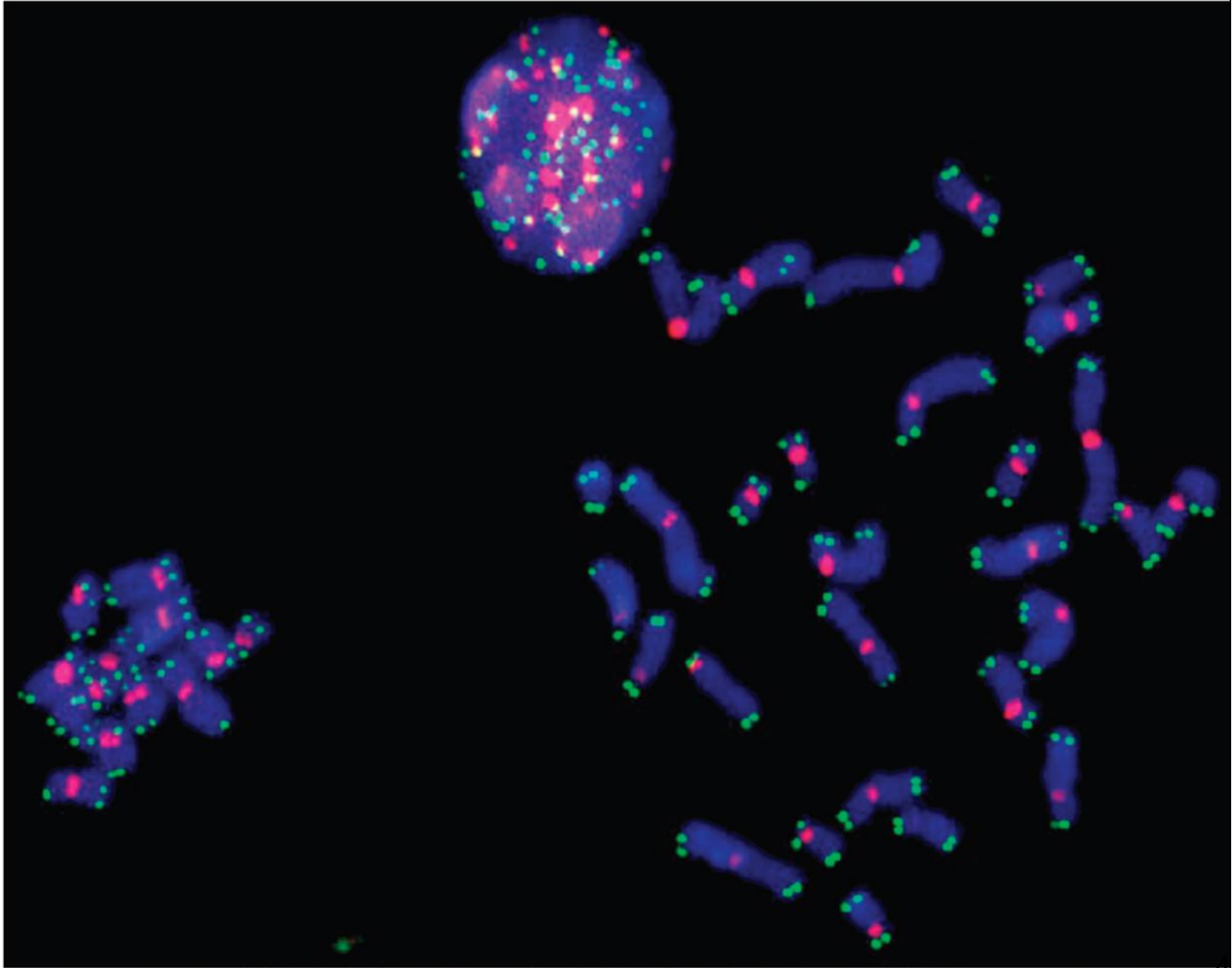


(Source: (c) Greider, C.W. and

Gaps left on both strand after replication



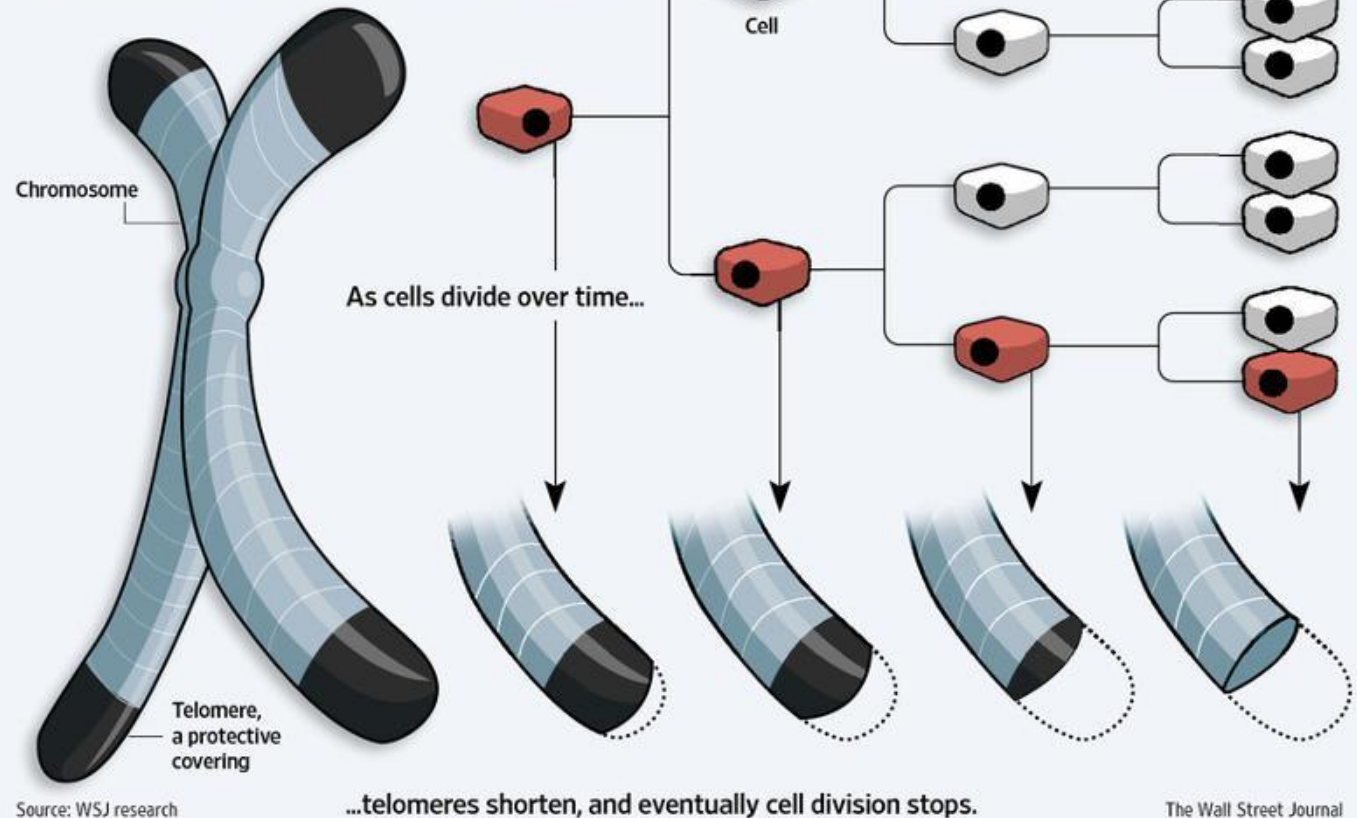




@ In human blood cells, the length of telomeres ranges from 8,000 base pairs at birth to 3,000 base pairs as people age and as low as 1,500 in elderly people. (An entire chromosome has about 150 million base pairs.) Each time a cell divides, an average person loses 30 to 200 base pairs from the ends of that cell's telomeres.

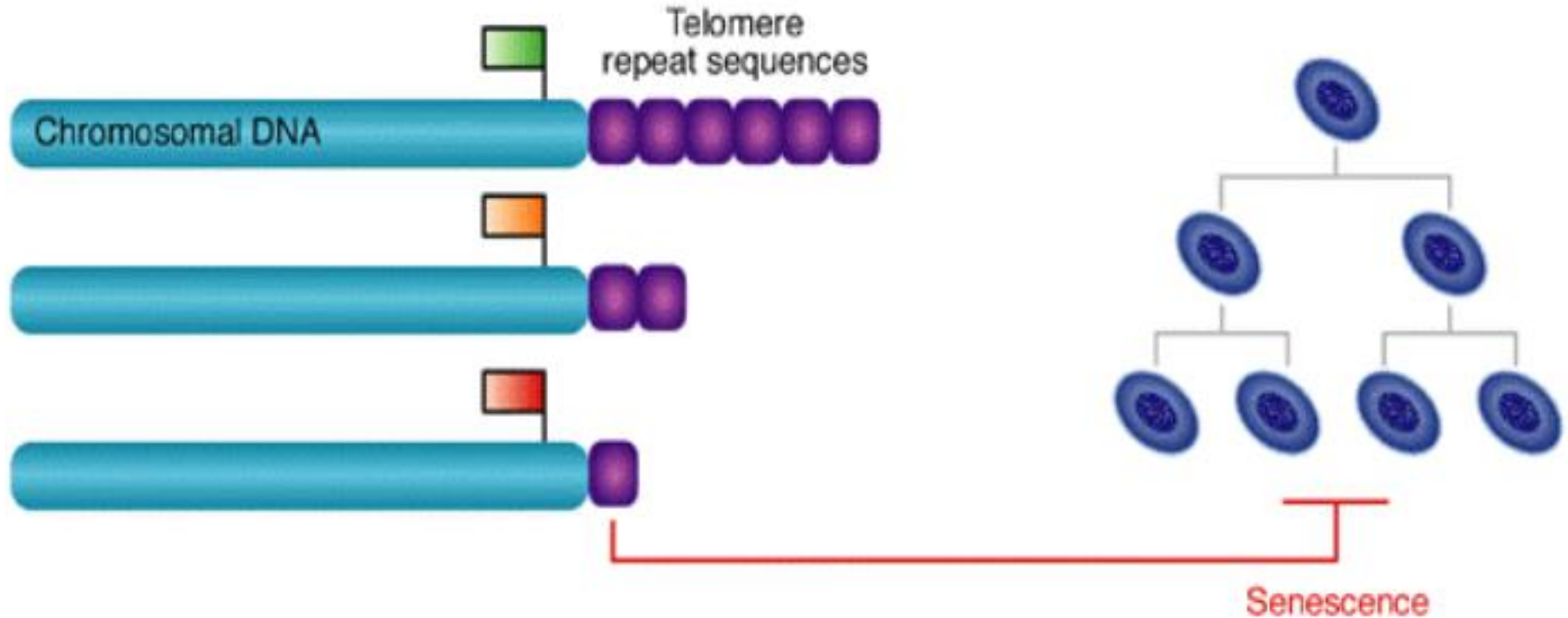
What We Lose With Age

As we grow older, telomeres at the end of our chromosomes shrink. New research suggests major depression also is linked to shorter telomeres, a sign of 'accelerated aging.'



Telomerase and Senescence

a Normal somatic cells



In most somatic tissues, telomerase is expressed at very low levels or not at all -- as cells divide, telomeres shorten. Short telomeres may be a signal for cells to senesce (stop dividing).

Summary of a typical prokaryotic replication fork (1)

1. Replication fork moves down the DNA as the helix is unwound catalyzed by **helicase** and **topoisomerase**. Re-annealing of the separated DNA strands is prevented by the acting of **SSBP**.

在拓扑异构酶与解链酶作用下，复制叉沿复制方向移动。单链结合蛋白结合在单链上保证单链不复性。

2. Both leading and lagging strands are synthesized by the dimer of **DNA pol III**. Continuous replication is going on for leading strand using the 3'-5' single strand as template.

前导链与滞后链均由DNA聚合酶III合成。前导链合成以3'—5'链为模板（按5'—3'合成），以引物合成酶产生的RNA为引物。

Summary of a typical prokaryotic replication fork (2)

3. **The lagging strand is looped around the DNA polymerase III enzyme, which allows one large pol III complex consisting of a pair of subunits to polymerize both leading and lagging strands simultaneously, then replication of both strands is thereby precisely coordinated .**

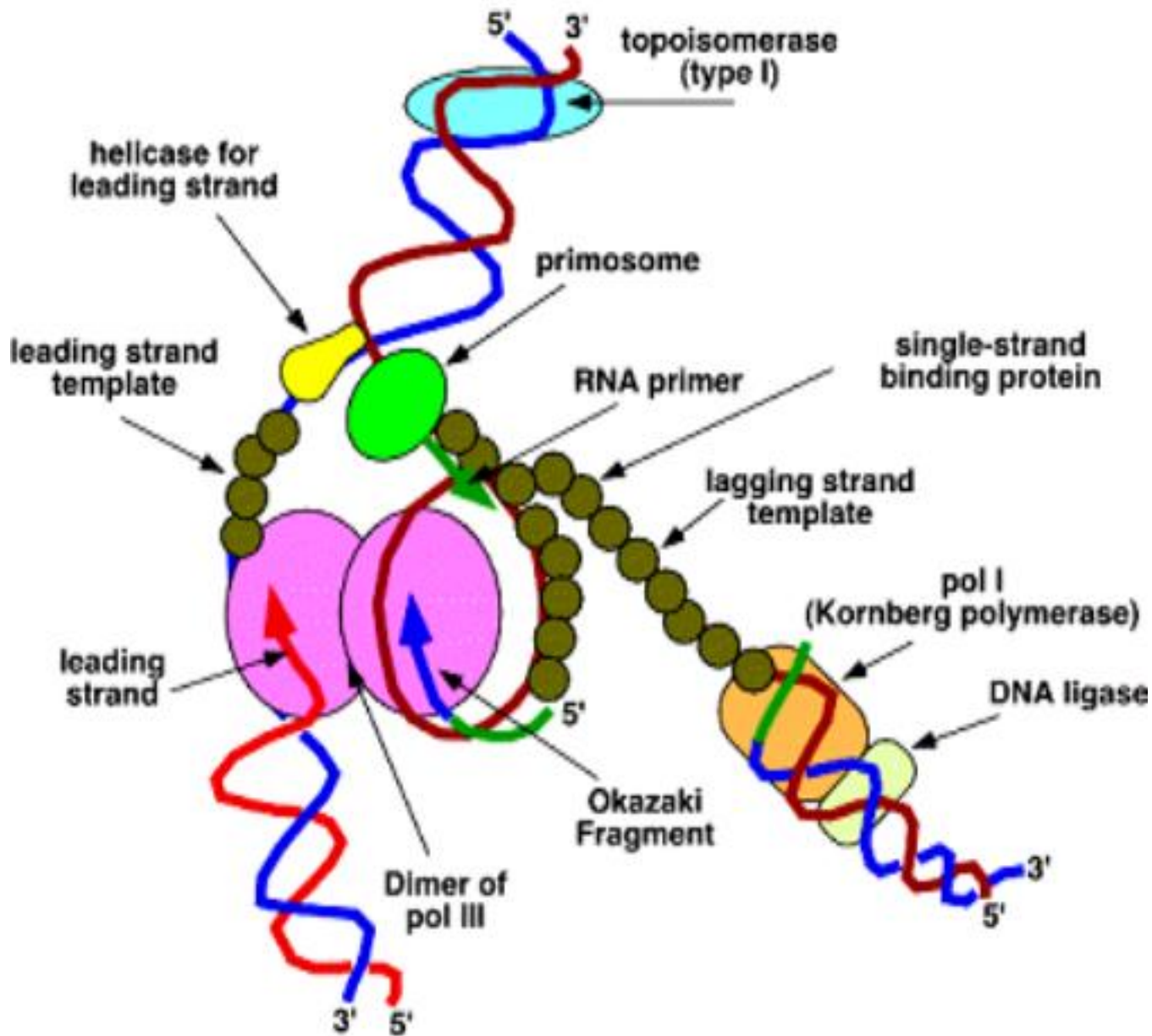
滞后链环绕在聚合酶III上使其合成方向也按前导链方向进行。同时也使其的合成与前导链合成协同。

Summary of a typical prokaryotic replication fork (3)

4. Primosome moves down the lagging strand and discontinuously produces RNA primer. Okazaki fragment (DNA) is produced using this RNA primer by DNA pol III. When the DNA fragment butts up the next Okazaki fragment, RNA primer is digested by RNaseH and the nick is sealed by DNA Pol I. The gap between Okazaki fragments is sealed by DNA ligase.

引物合成体在滞后链分叉的方向上前进，并在模板上断断续续地引发生成滞后链的引物RNA，再由DNA聚合酶III作用合成DNA（Okazaki片段），直至遇到下一个Okazaki片段，RNaseH与DNA聚合酶I均可降解RNA引物，并由DNA聚合酶I将缺口补齐，再由DNA连接酶将两个Okazaki片段连接形成大片段DNA。

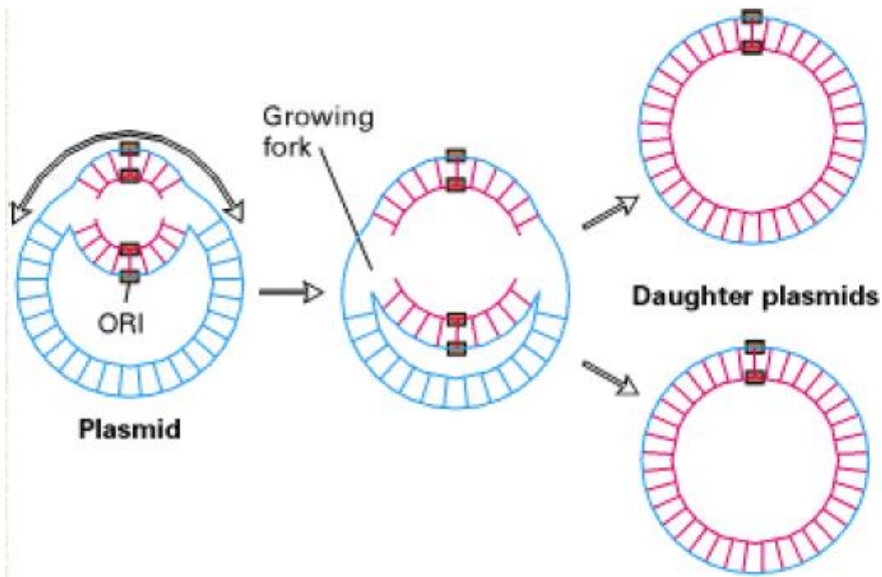
Replication Complex & Replication Fork



Ways of DNA replication

- **Bidirectional replication (双向复制):** Two replicating forks moving in opposite directions away from the origin.
- **Unidirectional replication(单向复制):** with one replicating fork move from the other
- **Rolling circle replication(滚环复制):** one strand of a double-stranded circular DNA remains intact & serves as the template for elongation of the other strand at a nick.

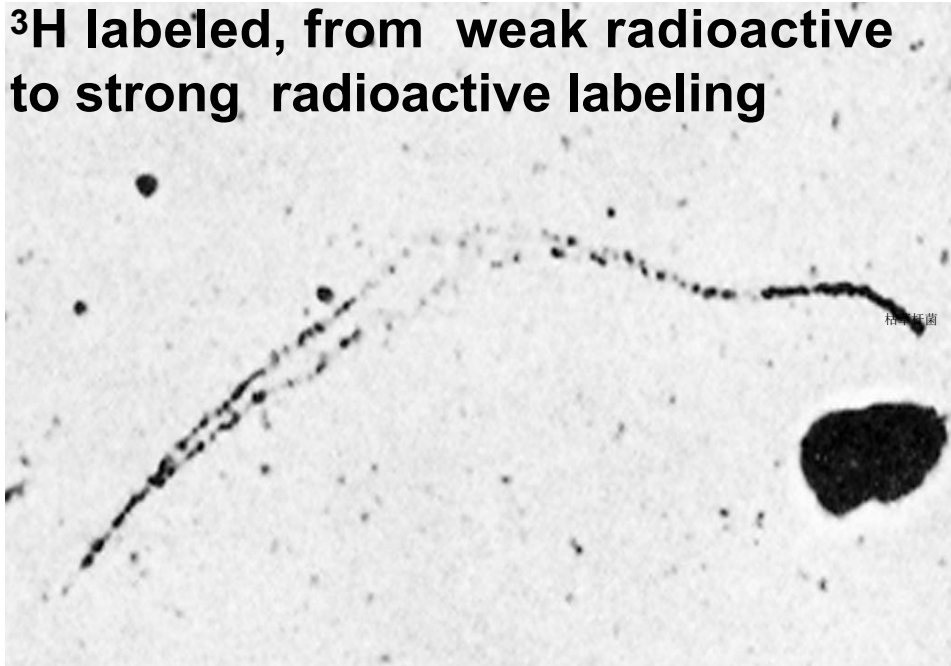
Bidirectional DNA Replication in Prokaryotes



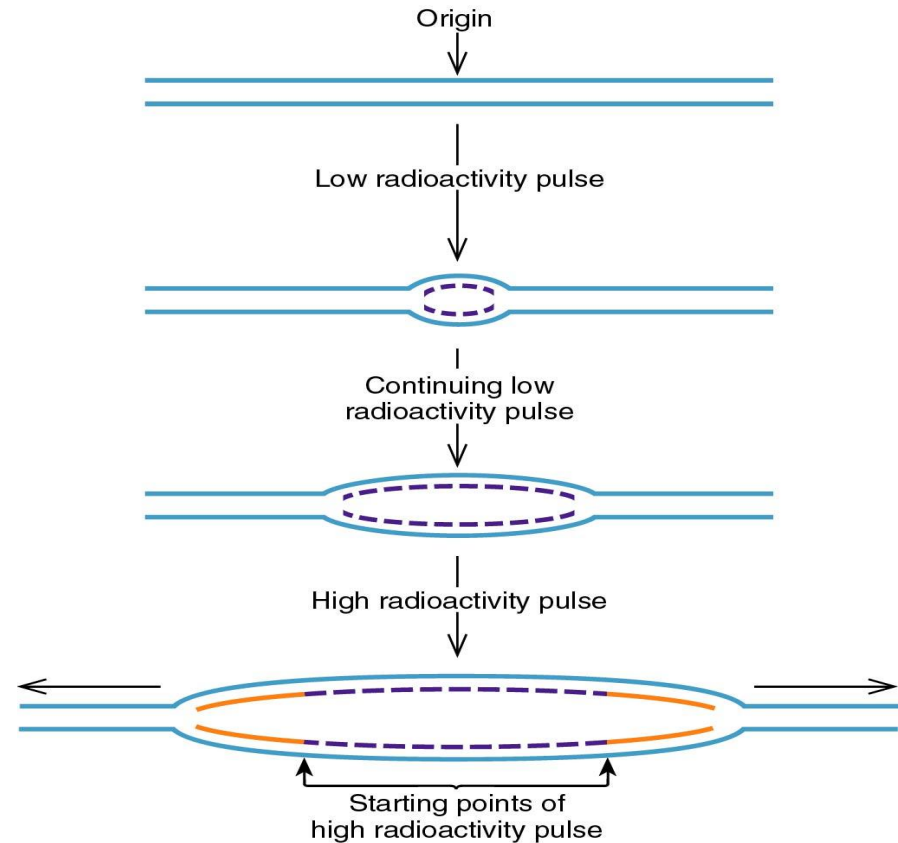
- θ structure contains two replicating forks

Bidirectional DNA Replication in prokaryote

^3H labeled, from weak radioactive to strong radioactive labeling



枯草杆菌 (*Bacillus subtilis*)



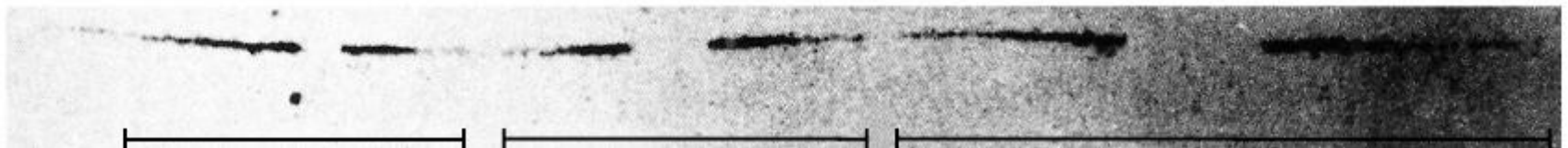
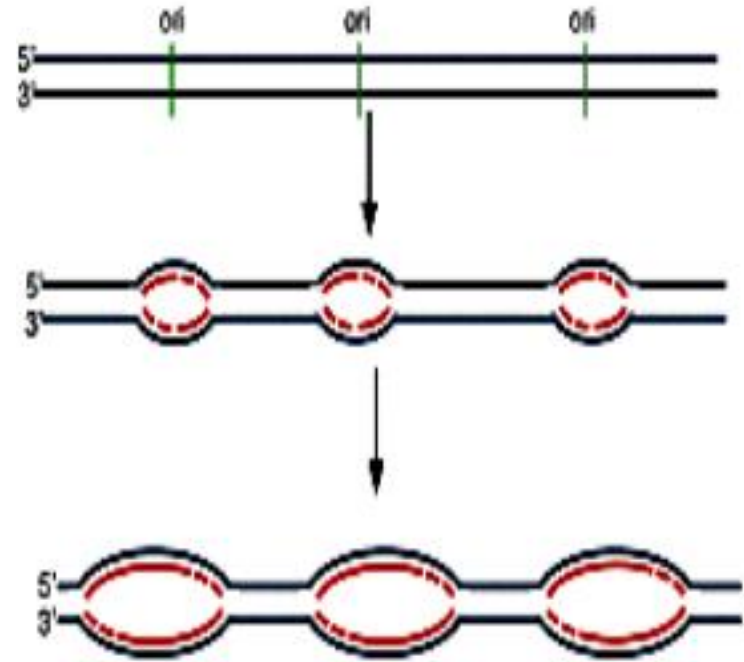
- Prokaryotic chromosomes are circular. DNA replication begins at a single origin point and proceeds around the circle in both directions

Origin of replication (ORI) & Replicon

- ORI(复制起始子): a unique DNA sequence at which DNA replication is initiated.
- Replicon (复制子) : all the DNA replicated from one origin of replication

Bidirectional DNA Replication in Eukaryotes

- In eukaryotic cells, the chromosomes are linear. They are also much larger. In order to keep replication within a reasonable time frame, these large chromosomes begin replication at several origin points. Each replication origin proceeds bi-directionally, again with leading and lagging strands according to the same rules.

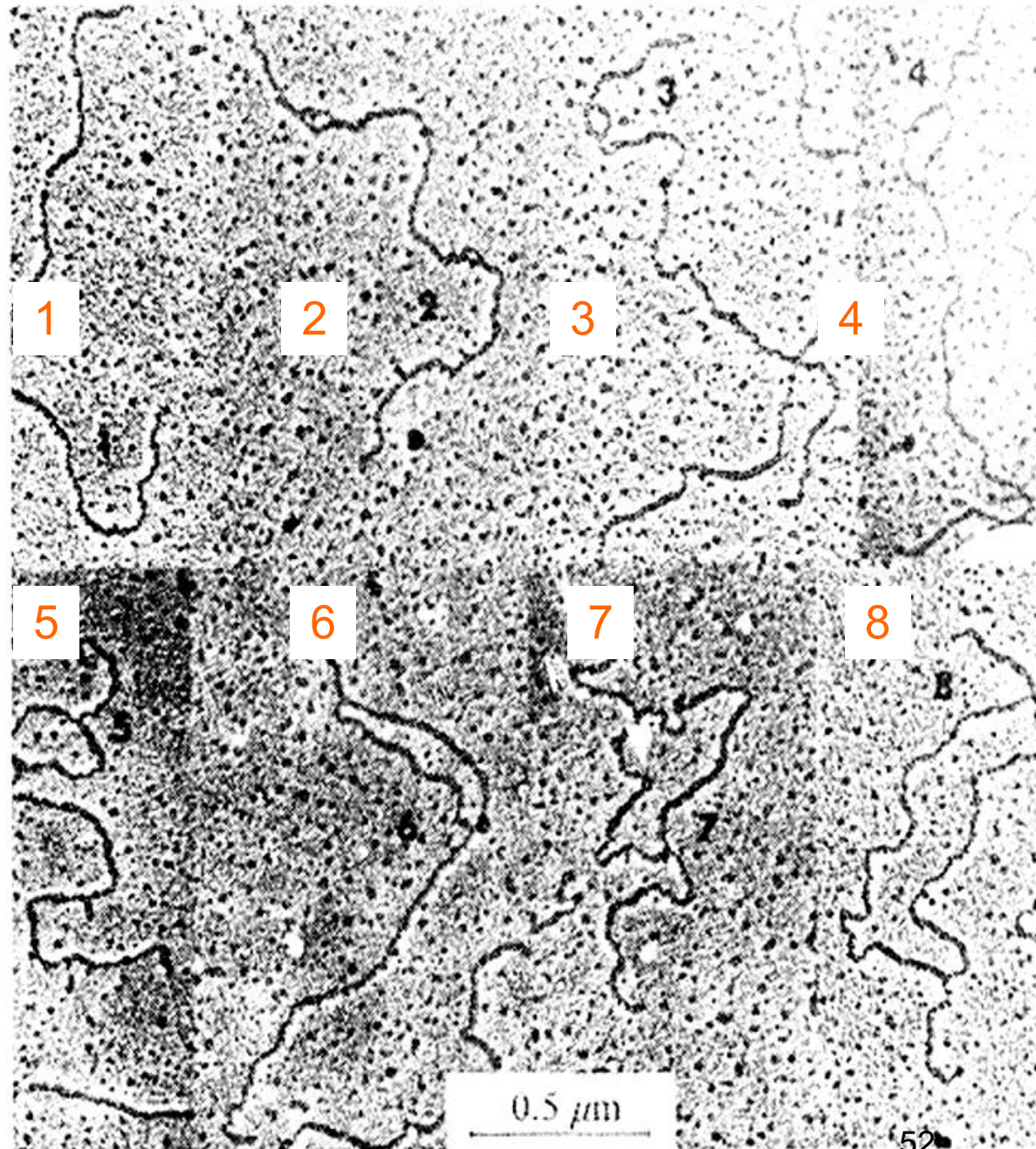


^3H labeled, from strong radioactive to weak radioactive labeling

Unidirectional replication

E. Coli 's plasmid colE1

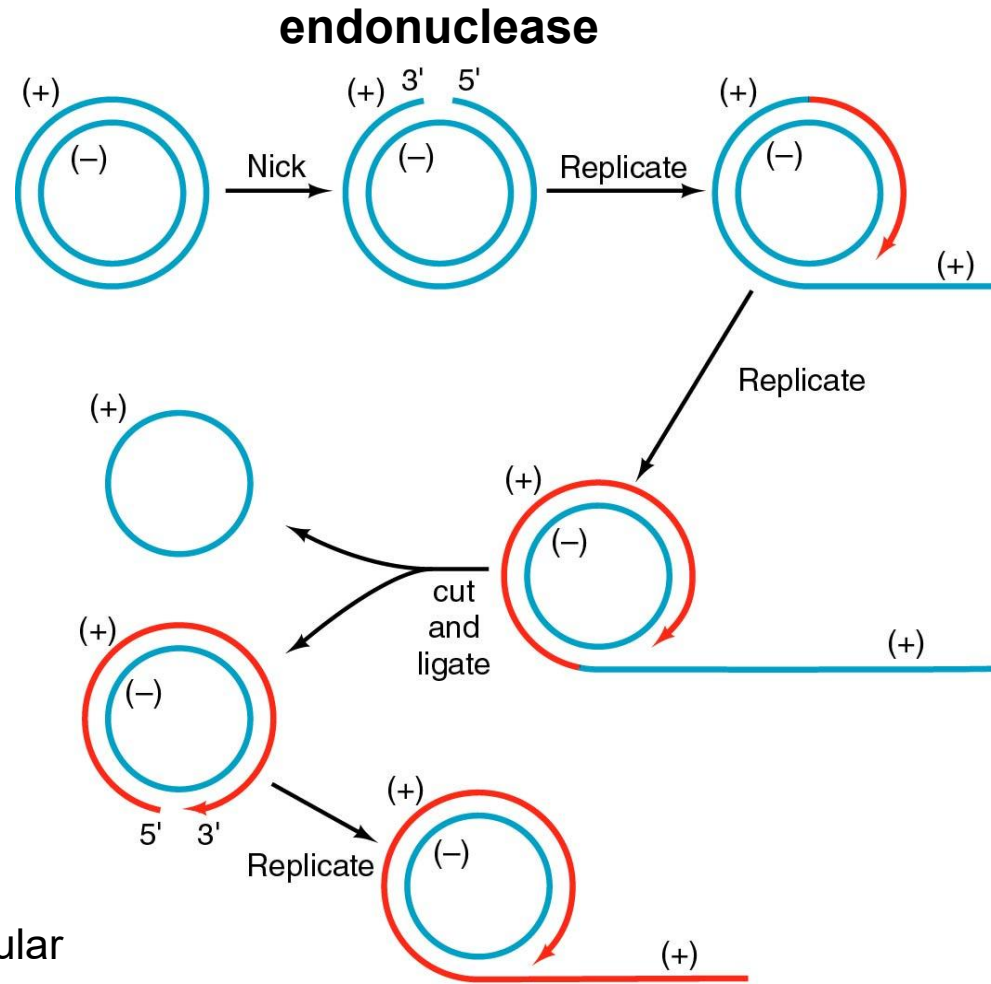
- Contains one ORI
- Contains a *EcoRI* site
- Using linerized plasmid DNA as template, it can be seen that the replication takes place in unidirectional.



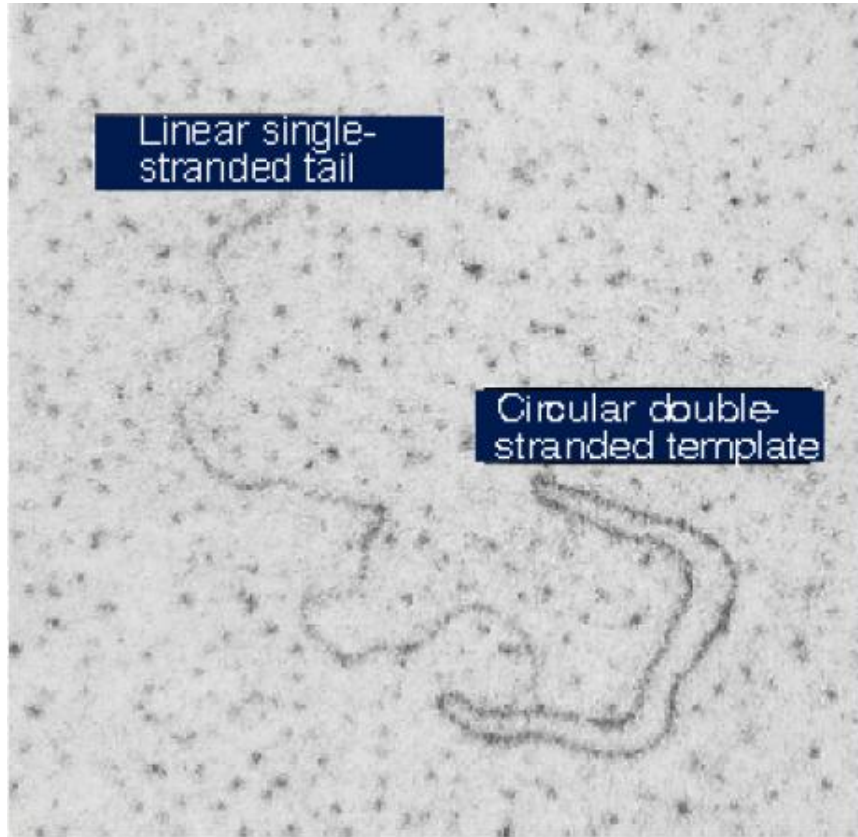
Rolling Circle replication

- Circular DNA can replicate by a rolling circle mechanism: One strand of a double-stranded DNA is nicked and 3'-end is extended, using the intact DNA as template.
- When one round of replication is complete, a full length, single-stranded circle of DNA is released
- It is called as σ model

E.coli phages with single-stranded circular DNA genomes, such as Φ X174



Rolling circles produce multimers of a replicon



A rolling circle appears as a circular molecule with a linear tail by electron microscopy

谢谢大家！